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Hearing protectors - Recommendations for selection, use, care and maintenance - Guidance document

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December 2025

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Protecteurs individuels contre le bruit - Recommandations relatives à la sélection, à l'utilisation, aux précautions d'emploi et à l'entretien - Document guide

Gehörschützer - Empfehlungen für Auswahl, Einsatz, Pflege und Instandhaltung - Leitfaden

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European foreword

This document (EN 458:2025) has been prepared by Technical Committee CEN/TC 159 “Hearing protectors” the secretariat of which is held by DIN.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by June 2026, and conflicting national standards shall be withdrawn at the latest by June 2026.

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. CEN shall not be held responsible for identifying any or all such patent rights.

This document supersedes EN 458:2016.

The main changes compared to the previous edition are listed in Annex I, Table I.1.

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Introduction

This document is intended to guide employers, supervisors and safety advisors. Additionally, this document gives information to all who need to use hearing protectors.

Hearing protectors are intended to protect against the risk of harmful noise and are elevated into the highest risk Category III under the Personal Protective Equipment Regulation (EU) 2016/425, which can cause irreversible health effects or can be potentially life threatening. They are intended to reduce the harmful effects that sound and noise can have on the hearing of an individual. Requirements for hearing protection products are given in the EN 352 series.

Guidance is provided on how to best select, use and maintain hearing protectors and check for compatibility for other equipment (e.g. Personal Protective Equipment (PPE), communication devices and hearing aids). Calculation procedures to estimate the residual noise exposure level, when a certain hearing protector is used, are also provided.

National bodies can develop their own local application documents based on this document.

Hearing protectors are generally available in two main forms: earmuffs and earplugs. Both forms are available with additional features and functions. All have their advantages and disadvantages in terms of attenuation, comfort, ease of use, communication facilities and cost.

In hearing conservation programmes noise hazard areas are identified and the personal noise exposure is assessed. Before a suitable hearing protector is considered, priority should be given to reducing noise at source and/or reducing the exposure time.

If the use of a hearing protector is found necessary or advisable, choosing optimum devices is a complex task. The most important aspects for hearing protector performance include sufficient attenuation and usage during the entire exposure period.

It is often desirable to retain the ability to hear speech and warning signals. If the hearing protector is a passive device only, the hearing protector should not provide excessive attenuation (over-protection), which can create a feeling of isolation and difficulties with perception of important sounds. In particular, this needs attention at moderate noise levels.

Hearing protectors are supplied with attenuation data in various formats. The attenuation is expressed in decibels and has been derived from laboratory tests. It is important to note that these data have been achieved under controlled laboratory conditions using trained test subjects. Under real working conditions, the attenuation achieved by the user can be lower than that generated by the laboratory testing.

The performance of hearing protectors is subject to natural variability in the anatomy amongst users. Correct fitting, training, regular inspection and user motivation are important to obtain the desired protection. Due to the natural variability, it is not possible to calculate the exact attenuation that a certain hearing protector will give for an individual. If a more accurate prediction is required, an individual fit test is strongly recommended. At high noise level exposures it is advisable to seek expert advice. In some cases dual protection, i.e. the use of an earmuff and an earplug combination, can be required.

For hearing protectors to be effective they should be used at all times when the user is in a potentially hazardous noise environment. When selecting hearing protectors, attention should be given to factors influencing comfort and user preference.

This document also provides advice on how to consider sustainability.

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1 Scope

This document gives recommendations for the selection, use, care and maintenance of hearing protectors.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

EN 352-6, *Hearing protectors — Safety requirements — Part 6: Earmuffs with safety-related audio input*

EN 352-8, *Hearing protectors — Safety requirements — Part 8: Entertainment audio earmuffs*

EN 352-9, *Hearing protectors — Safety requirements — Part 9: Earplugs with safety-related audio input*

EN 352-10, *Hearing protectors — Safety requirements — Part 10: Entertainment audio earplugs*

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminology databases for use in standardization at the following addresses:

— ISO Online browsing platform: available at https://www.iso.org/obp/

— IEC Electropedia: available at https://www.electropedia.org/

3.1
daily noise exposure level
$L_{\mathrm{EX,8h}}$
A-weighted noise exposure level normalized to a nominal 8 h working day according to EN ISO 9612

3.2
peak sound pressure level
$L_{p,\mathrm{Cpeak}}$
C-weighted instantaneous peak sound pressure level according to EN ISO 9612

3.3
national regulation level
L'_{NR}
daily noise exposure level ($L_{\mathrm{EX,8h}}$) effective to the ear according to national regulations

Note 1 to entry: National laws or regulations determine the levels of exposure at which hearing protectors are provided or used.

3.4
national peak regulation level
$L'_{\mathrm{NR,peak}}$
peak pressure level $L_{p,\mathrm{Cpeak}}$ effective to the ear according to national regulations

Note 1 to entry: National laws or regulations determine the levels of exposure at which hearing protectors are provided or used.

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3.5
effective attenuation
measure of protection afforded by the hearing protector for the user

3.6
A-weighted sound pressure level
$L_{p,\mathrm{A}}$
sound pressure level with frequency weighting A according to EN 61672-1

3.7
A-weighted sound pressure level effective to the ear
$L'_{p,\mathrm{A}}$
A-weighted diffuse-field related sound pressure level under the hearing protector for the external sound pressure level $L_{p,\mathrm{A}}$

3.8
C-weighted sound pressure level
$L_{p,\mathrm{C}}$
sound pressure level with frequency weighting C according to EN 61672-1

3.9
A-weighted equivalent sound pressure level effective to the ear
$L'_{p,\mathrm{A,eq}T}$
A-weighted diffuse-field related equivalent sound pressure level under the hearing protector for the external equivalent sound pressure level for time period T, $L_{p,\mathrm{A,eq}T}$

Note 1 to entry: For simplicity of notation, the subscript T is omitted throughout the following text.

3.10
daily noise exposure level effective to the ear
$L'_{\mathrm{EX,8h}}$
A-weighted diffuse-field related equivalent sound pressure level under the hearing protector for the external daily noise exposure level $L_{\mathrm{EX,8h}}$

3.11
peak sound pressure level effective to the ear
$L'_{p,\mathrm{Cpeak}}$
C-weighted peak sound pressure level under the hearing protector for the external sound pressure level $L_{p,\mathrm{Cpeak}}$

3.12
flat frequency response
attenuation which is constant (or nearly constant) over the frequency range between 125 Hz and 8 000 Hz

Note 1 to entry: The slope of the linear regression of mean attenuation values from 125 Hz up to 4 000 Hz fulfils the criterion proposed by Liedtke [25] with a value lower than 3,6 dB, or very approximately $H - L < 9$ dB. See A.1 for explanation of "H" and "L".

3.13
fit testing
<hearing protectors> procedure for checking that a specific hearing protector is suitable for use by a specific individual by assessing the physical fit, seal, sound attenuation or other properties of the hearing protector

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[SOURCE: EN 17479:2021, 3.1]

3.14
personal attenuation rating
PAR
individual attenuation given as a single value in dB that a user obtains for the fit of the hearing protector that was tested

Note 1 to entry: The PAR can be either the combined left-right (binaural) or separate left-right ear value.

Note 2 to entry: The calculation procedure is not standardized and is specified by the manufacturer for a specific fit testing method.

[SOURCE: EN 17479:2021, 3.4]

4 Symbols and abbreviations

For the purposes of this document, the following symbols and abbreviations apply.

f	centre frequency of the octave band in Hz
$L_{p,\mathrm{eq}}$	equivalent sound pressure level
$L_{p,\mathrm{oct}}$	sound pressure level for a single octave band
$L_{p,\mathrm{oct,eq}}$	equivalent sound pressure level for a single octave band
PNR	predicted noise level reduction according to EN ISO 4869-2
APV_f	assumed protection value APV_{f84} according to EN ISO 4869-2
H	high-frequency attenuation value
M	middle-frequency attenuation value
L	low-frequency attenuation value
SNR	Single Number Rating
rms	root mean square
$L_{\mathrm{EX,8h}}$	daily noise exposure level
$L_{p,\mathrm{Cpeak}}$	C-weighted peak sound pressure level
L'_{NR}	national regulation level
$L'_{\mathrm{NR,peak}}$	national peak regulation level
$L_{p,\mathrm{A}}$	A-weighted sound pressure level
$L'_{p,\mathrm{A}}$	A-weighted sound pressure level effective to the ear
$L_{p,\mathrm{C}}$	C-weighted sound pressure level
$L_{p,\mathrm{A,eq}T}$	A-weighted equivalent sound pressure level
$L_{p,\mathrm{C,eq}T}$	C-weighted equivalent sound pressure level
$L'_{p,\mathrm{A,eq}T}$	A-weighted equivalent sound pressure level effective to the ear

NOTE For simplification of notation, the subscript T is omitted throughout the following text.

$L'_{\mathrm{EX,8h}}$	daily noise exposure level effective to the ear

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$L'_{p,\mathrm{Cpeak}}$	peak sound pressure level effective to the ear
$L_{\mathrm{C,Fmax}}$	maximum value of the sound pressure level with time-weighting F (fast) and frequency weighting C
$L_{\mathrm{A,Fmax}}$	maximum value of the sound pressure level with time-weighting F (fast) and frequency weighting A
A_f	frequency weighting A in dB for octave band centre frequency f
L_{crit}	criterion level
L_{max}	maximum input signal level to the protector audio input
U_{crit}	criterion voltage
U_{max}	maximum rms voltage input to the protector audio input
K	constant for corresponding national regulation level
n	maximum number of hours for use of audio input
PPE	personal protective equipment
ANR	active noise reduction
REAT	real ear attenuation at threshold
d_{m}	modified sound attenuation value

5 Types of hearing protectors

5.1 Design forms

5.1.1 Earmuffs

Earmuffs consist of cups which fit over the ears and are sealed to the head with soft cushions, usually filled with foam and/or liquid. The cups are usually lined with sound absorptive material. They are connected by a tensioning band (head band), usually made of metal and/or plastic. When the earmuff is used behind-the-head or under-the-chin and has a mass in excess of 150 g, a flexible head strap is fitted to each cup or to the head band close to the cups to support the earmuff. Some earmuffs have one cup intended only for the left ear and the other only for the right ear. Earmuffs can be available in ‘medium size range’, ‘small size range’ and ‘large size range’ types. ‘Medium size range' earmuffs will fit the majority of users. ‘Small size range' or ‘large size range' earmuffs are designed to fit users for whom ‘medium size range' earmuffs are not suitable.

Earmuffs are available with head bands, neck bands, chin bands and universal bands. Earmuffs with neck bands and chin bands permit the simultaneous using of a safety helmet. Universal bands can be used over-the-head, behind-the-head or under-the-chin.

Requirements for earmuffs are specified in EN 352-1.

5.1.2 Earmuffs attached to head protection and/or face protection devices

Earmuffs attached to head protection and/or face protection devices (also called mounted earmuffs) consist of individual cups attached to arms that are mounted to a safety helmet or other equipment serving as carrier for the hearing protector, such as the rigid head top of a powered and supplied air respirator system. The arms are adjustable so that the cups can be positioned over the ears.

Requirements for earmuffs attached to head protection and/or face protection devices are specified in EN 352-3.

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5.1.3 Earplugs

5.1.3.1 General

Earplugs are hearing protectors that are designed to be inserted into the ear canal or to cover the ear canal entrance. They are sometimes provided with an interconnecting cord (corded earplugs), rigid band or finger grips. Earplugs can be either disposable (intended for single use) or reusable (intended for repeated use). All types of earplugs except custom moulded earplugs can be available in a range of sizes.

Requirements for earplugs are specified in EN 352-2.

5.1.3.2 Pre-shaped earplugs

Pre-shaped earplugs can readily be inserted into the ear canal without prior shaping. Pre-shaped earplugs are available in a variety of materials.

5.1.3.3 User-formable earplugs

User-formable earplugs are made from compressible or malleable materials that the user forms before insertion. Compressible earplugs are designed to expand to form the seal within the ear canal after insertion.

5.1.3.4 Banded earplugs

Banded earplugs are pre-shaped earplugs attached to a band which presses them into the ear canal or against the entrance of the ear canal. Some are intended to be used in more than one position, e.g. with the band under-the-chin.

5.1.3.5 Custom moulded earplugs

Custom moulded earplugs are individually moulded to fit the shape of the user's ear canals typically using an impression or a scan of the concha and the ear canal of the user. They can be manufactured in a range of materials and with different acoustic filters to offer a range of attenuation.

5.2 Function mode

5.2.1 Basic function mode

All hearing protectors have the characteristics of reducing noise by their design and type of material used, to absorb and/or reflect sound. Devices that have only this function are called passive devices. They constitute the basis for all other additional function modes as described in 5.2.2 to 5.2.4.

5.2.2 Level-dependent hearing protectors

5.2.2.1 General

Level-dependent hearing protectors are designed to provide different attenuation as the external sound level changes. Their main purpose is to protect against impulsive or intermittent hazardous noise while allowing situational awareness.

5.2.2.2 Passive level-dependent hearing protectors

Passive level-dependent hearing protectors use the acoustic properties of carefully designed air ducts to give different attenuation depending on the noise level. These types of hearing protectors are designed to be effective against very high level single-impulse noises, such as firearms, rather than the continuous noise or repetitive impulses found in most industrial situations.

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5.2.2.3 Sound-restoration level-dependent hearing protectors

Sound-restoration level-dependent hearing protectors incorporate an electronic sound reproduction system. At low sound pressure levels, the sound detected by an external microphone is relayed to a loudspeaker inside the earmuff or earplug. As the external sound pressure level increases, the electronics reduce the gain and control the level of reproduced sound inside the hearing protector.

Requirements for level-dependent hearing protectors are specified in EN 352-4 (for earmuffs) and EN 352-7 (for earplugs), e.g. minimum criterion levels.

There are two types of sound-restoration level-dependent hearing protectors: products only intended for use in impulsive type noise situations and products for all types of noise environment (continuous, fluctuating and intermittent noise including impulsive noise, see Annex G).

5.2.3 Active noise reduction (ANR) protectors

Active noise reduction protectors are hearing protectors which incorporate an electronic sound cancelling system to achieve additional noise attenuation where passive hearing protectors can be less effective. ANR is particularly effective for continuous noise at low frequencies (50 Hz to 500 Hz) (see Annex G).

Requirements for active noise reduction earmuffs are specified in EN 352-5.

5.2.4 Hearing protectors with external audio input

5.2.4.1 General

These devices use a wired or wireless system through which communication signals, alarms, messages or audio entertainment can be relayed. Some products incorporate a system to limit the sound pressure level.

5.2.4.2 Hearing protector with entertainment audio input

These devices can incorporate a radio receiver set or music player for entertainment or allow audio input from external devices. The entertainment audio input is limited to 82 dB(A).

Requirements for entertainment audio hearing protectors are specified in EN 352-8 (for earmuffs) and EN 352-10 (for earplugs).

5.2.4.3 Hearing protector with safety-related audio input

These devices include wired or wireless input and/or built-in two-way radio and can be used for work-related or safety-related communication. As the information to be received might be crucial for safety, the product standards impose no limitation on the reproduced sound pressure level at the ear.

Requirements for hearing protectors with safety-related audio input (for work or safety-related communication) are specified in EN 352-6 (for earmuffs) and EN 352-9 (for earplugs).

6 Selection

6.1 Principles

Personal hearing protection should be selected so that, when used correctly and for the entire duration of exposure, it will eliminate or minimize the risk to hearing.

As there are many different hearing protectors intended for use in a wide range of noise environments it is important to choose a suitable type. The product shall be checked for regulatory conformance. Consideration should be given to the factors listed in a) to h). The list is neither exclusive nor exhaustive:

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a) sound attenuation, see 6.2;

b) work environment, see 6.3;

c) essential work or safety-related communication, especially speech intelligibility, see 6.4;

d) compatibility with other personal protective equipment (PPE), such as helmets, spectacles, etc., see 6.5;

e) the interaction between the user and the situation of use, see 6.6;

f) medical factors, see 6.7;

g) user comfort and ergonomic requirements, see 6.8;

h) sustainability aspects of the products, see Clause 9.

The selection procedure should be reviewed at regular intervals to ensure that an effective attenuation is maintained. When considering all factors in the selection process the most important outcome is that the hearing protector will be used during the whole time of noise exposure.

6.2 Selection according to sound attenuation

6.2.1 Guide to protection rating

Hearing protectors should be chosen according to the sound attenuation they will provide. Methods to predict the sound attenuation of hearing protectors are given in 6.2.3.

NOTE 1 It is generally accepted that the risk of hearing damage associated with occupational noise exposure is low where the daily noise exposure level ($L_{EX,8h}$) is below 80 dB, and insignificant where $L_{EX,8h}$ is below 75 dB.

NOTE 2 National regulations or other guidelines can stipulate selection criteria for personal hearing protection and place limits on sound exposure. Such relevant criteria and limits are often taken into account in deciding what level of sound attenuation is required.

In general, a hearing protector which provides an effective sound level at the ear ($L'_{p,A,eq}$) of between 70 dB and 80 dB is considered suitable. If the chosen hearing protector provides excessive attenuation, users are at risk of failing to recognize warning signals and understand essential communications. Users can also feel isolated from their working environment. The perception of isolation increases as the sound level effective to the ear decreases.

Workers can be exposed to different noise environments during the working day. It can be possible to select a single hearing protector which is adequate for all situations likely to be encountered, or in some situations, to select more than one type of hearing protector.

EXAMPLE 1 A worker's only significant noise exposure is to an $L_{p,A,eq}$ of 98 dB, for a total of 30 min per day. His $L_{EX,8h}$ is 86 dB. His employer wishes to minimize the risk to the worker's hearing, and so aims to provide a hearing protector that provides an effective sound pressure level at the ear of below 80 dB, ideally between 75 dB and 70 dB during the noise exposure. Using the *HML* check method, the employer selects a protector that provides a sound attenuation of M = 26 dB.

EXAMPLE 2 A worker spends the majority of her working day (a total of 6 h) in an environment with an $L_{p,A,eq}$ of 94 dB. Her $L_{EX,8h}$ is around 93 dB. Her employer wishes to reduce the effective $L'_{EX,8h}$ to at least below 80 dB, and to minimize the risk to the worker's hearing. The employer rejects a protector that provides a sound attenuation of M = 32 dB, as this would reduce the effective sound pressure level at the ear to 62 dB and so risk excessive attenuation. Using the *HML* check method, the employer selects a protector that provides a sound attenuation of M = 22 dB.

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6.2.2 Sound attenuation in practice

The attenuation afforded by a hearing protector when used under normal working conditions can differ from that indicated on the device packaging or in the user information. This can be due to factors including incorrect selection of type and/or size, use (see Clause 7), misuse and poor maintenance. For remarks on improving field performance see Annex E.

These differences can be overcome by appropriate information, instruction and on-going training in the use of the product. There are tools, referred to as “fit testing systems” that allow individual tests to be carried out to verify the correct fit. Further information and guidance on these systems can be found in 7.3 and more extensively in EN 17479.

In the absence of fit testing, some countries have published specific guidance on de-rating the published data in an attempt to address the issue of attenuation in practice. Refer to national guidance for further information.

NOTE Sound attenuation in practice can be referred to as real-world attenuation or field attenuation.

6.2.3 Acoustical selection methods

6.2.3.1 Overview of applicable methods

The applicability of methods for estimating the sound pressure level effective to the ear for different types of hearing protector is summarized in Table 1. If the risk assessment indicates that the national peak regulation level is likely to be exceeded, refer also to the last column of Table 1.

Table 1 — Applicability of acoustical selection methods to different types of hearing protector

Type of hearing protector	Selection based only on L'_{NR}	Selection also based on $L'_{NR,peak}$
Passive	Annex A	Annex B
Passive level dependent	Annex A	Annex B
Sound restoration level dependent	Annex C	Annex B [a]
ANR	Annex A [b]	Annex B [a]
Audio input	Annex A and Annex D	Annex B [a]
Dual hearing protection	Annex A [c]	Annex B [c]

[a] using the passive attenuation data
[b] using the total (passive and active) attenuation data
[c] using the passive attenuation data for the combination

6.2.3.2 Methods for estimating the level effective to the ear when selecting a hearing protector based on the national regulation level L'_{NR}

This approach is suitable for continuous, fluctuating and intermittent noise.

When selecting a suitable hearing protector, consideration should be given to the characteristics of the noise and the attenuation data of potentially suitable hearing protectors. The attenuation of most hearing protectors varies with frequency. To determine if a hearing protector is (acoustically) suitable, it is necessary to estimate the sound pressure level effective to the ear when the hearing protector is used. There are four methods of estimating the sound pressure level effective to the ear. These are:

— Octave band method;

— *HML* method;

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— *HML* check method;

— *SNR* method.

All four methods are explained in detail in Annex A including worked examples. Table 2 shows the type of information on workplace noise needed for each estimation method.

Table 2 — Information on workplace noise required for assessing sound attenuation

Estimation method	Information required
Octave band method (see A.2)	octave band sound pressure level; $L_{p,\mathrm{oct}}$ / $L_{p,\mathrm{oct,eq}}$
HML method (see A.3)	A- and C-weighted sound pressure level; $L_{p,\mathrm{A}}$ and $L_{p,\mathrm{C}}$ / $L_{p,\mathrm{A,eq}}$ and $L_{p,\mathrm{C,eq}}$
HML check method (see A.4)	A-weighted sound pressure level; $L_{p,\mathrm{A}}$ / $L_{p,\mathrm{A,eq}}$ subjective decision between two noise classes (using lists of examples of noise sources, see A.4)
SNR method (see A.5)	C-weighted sound pressure level; $L_{p,\mathrm{C}}$ / $L_{p,\mathrm{C,eq}}$

For changing noise environments and/or exposure times it can be appropriate to use more than one type of hearing protector in a working day. If the noise is not continuous but fluctuating or impulsive, the equivalent sound pressure level should be calculated or measured. The four estimation methods do not take account of any de-rating values. Refer to national guidance for further information on this issue (see 6.2.2).

6.2.3.3 Method for estimating the level effective to the ear when selecting a hearing protector based on the national peak regulation level $L'_{\mathrm{NR,peak}}$

The method described in Annex B is applicable to all hearing protectors. At high peak sound pressure levels non-passive devices operate in the passive mode and therefore passive *HML* attenuation data are used.

6.2.3.4 Selection method for sound-restoration level-dependent hearing protectors intended for all types of noise environment, using *HML* criterion values

Annex C describes a method to aid the correct selection of a non-passive sound-restoration level-dependent hearing protector for use in all types of noise environment. The method is applicable to both earmuffs and earplugs using *HML* criterion level data obtained in accordance with EN 352-4 and EN 352-7 respectively. The method can be suitable for other similar non-passive devices. The criterion levels describe the maximum external noise level for which this product does not lead to more than 85 dB(A) effective to the ear. The criterion levels are determined with volume control set to maximum.

The annex determines only whether the hearing protector will reduce the A-weighted sound pressure level effective to the ear to below the defined limit of 85 dB(A). This annex cannot give the specific level at the ear.

6.2.3.5 Method for estimating the level effective to the ear for an active noise reduction hearing protector based on the national regulation level L'_{NR}

The selection can be made following the methods of Annex A according to the octave band method in A.2, the *HML* method in A.3 or the *HML* check method in A.4, using the total attenuation values according to EN 352-5. In addition, it should be checked that the external sound pressure level for which the sound pressure level effective to the ear remains linear is not exceeded (these data are obtained from the instructions for use). The outcome of the assessment is to be compared to the national regulation level L'_{NR}. The product should be assessed based on the passive attenuation values in the event that the active noise reduction is not operational.

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6.2.3.6 Method for estimating the level effective to the ear for a hearing protector with audio input

When there is a need or demand for radio for essential communication or entertainment, hearing protectors with an audio input are available. These are divided in two types: for entertainment (e.g. listening to a music source) or for safety or work-related speech communication.

The user will be exposed to both the attenuated ambient noise and the audio input signal. Hearing protectors with entertainment audio input shall conform to EN 352-8 (for earmuffs) or EN 352-10 (for earplugs). For products with wireless or wired entertainment audio input the sound pressure level is limited to 82 dB ($L'_{p,\mathrm{A,eq}}$).

For hearing protectors with electrical audio input the sound pressure level can exceed 82 dB(A) for input voltages higher than those specified by the manufacturer. To minimize the risk of hazardous exposure it is therefore recommended to select products with built-in limiter functions.

For essential communication, hearing protectors shall conform to EN 352-6 (for earmuffs) or EN 352-9 (for earplugs). The following information is provided by the manufacturer:

— the input signal level corresponding to 82 dB(A) effective to the ear (criterion input signal);

— the relationship between sound output level and input signal in 5 dB steps from a sound output level of 70 dB(A) up to the maximum input signal;

— the sound output level for maximum input signal;

— the usage time for the maximum input signal that corresponds to an equivalent sound output level of 82 dB(A) over 8 h.

NOTE The usage time can be increased, if the input signal level is lowered.

If the contribution of the audio input signal to the sound level effective to the ear is substantial the combined exposure should be considered. It can be necessary to reduce the contribution of the external source. Annex D describes a calculation method for combining these sources.

When selecting this type of product, it is important to realize that the additional sound input will negatively affect the awareness of the sound from the environment and warning signals. Systems providing the input signal to only one ear can be a safer solution. The audibility of warning signals should be checked.

6.2.4 Selection of combination of earmuffs and earplugs

Personnel working in extremely noisy environments can require higher attenuation than a single hearing protector can provide. In such cases, increased protection can be obtained by using a combination of earmuffs and earplugs. Dual protection should be considered when the daily noise exposure exceeds 105 dB(A) if there is substantial noise at frequencies below 500 Hz. For higher frequency dominant noise dual protection should be considered from 110 dB(A). The attenuation of a combination of earmuffs and earplugs is not the sum of the attenuation of the individual protectors. The typical increase in attenuation will depend on the noise, fit and what combination of protectors is used.

NOTE Results from 46 combination measurements show a range of 1 dB to 12 dB increase above the higher *SNR* (see [22]). For seven combinations of earmuffs with high attenuating earplugs an increase in *M*-value between 5 dB and 9 dB was observed. The *L*-value can vary in a wider range and is dependent strongly on the attenuation of the earplug.

Where dual protection is required, it is important to obtain attenuation data for the chosen combination of earplug and earmuff (and helmet, if used), e.g. by contacting the manufacturer. If both the earplug and earmuff are well fitted, the increase in attenuation is mainly at frequencies below 1 000 Hz. Bone

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conduction can limit the total sound attenuation, particularly around 2 000 Hz. The most suitable combinations of earplug and earmuff (dual hearing protection) are dependent on both the noise as well as any need for communication, situational awareness and issues related to use of personal protective equipment.

6.3 Selection according to work environment

6.3.1 Factors of work environment – overview

Further to the acoustical selection guidance (see 6.2) consideration should also be given to influences present in the work environment. These can include, but are not limited to, temperature, dust, hazardous substances, moving machinery, vibration, localization of noise sources, feeling of isolation and the daily time of use of the hearing protector.

In addition, the importance of hearing warning signals and informative sounds should also be taken into consideration (see 6.3.2.8).

6.3.2 Acoustical factors

6.3.2.1 Influence of types of noise

The selection of hearing protectors should consider the type(s) of noise in the working environment:

a) continuous noise;

b) fluctuating noise;

c) intermittent or repeated short-term noise;

d) impulsive noise;

e) dominant low frequency noise;

f) high frequency pure tones in the noise (increased risk of hearing damage);

g) presence of informative sounds.

For continuous, fluctuating or intermittent noise the selection is determined by the daily noise exposure level ($L_{\mathrm{EX,8h}}$). For impulsive noise the selection is determined by the daily noise exposure level ($L_{\mathrm{EX,8h}}$) and by the peak sound pressure level ($L_{p,\mathrm{Cpeak}}$).

See Annex G for illustrations of the different noise types.

6.3.2.2 Exposure to continuous noise

In a continuous or steady-state noise environment, most types of hearing protectors can be used. Other influencing factors, such as communication requirements or environmental factors, can help determine which particular type of hearing protector is most suitable (see 6.3.3 and 6.4).

6.3.2.3 Exposure to fluctuating noise

In a fluctuating noise environment, a level-dependent or sound-restoration hearing protector can be more suitable. Passive or other hearing protectors can also be used subject to an appropriate risk assessment. In fluctuating noise, there is a risk that users will remove their hearing protection during prolonged periods of lower noise level. This is even more likely if the selected hearing protector is providing excessive attenuation. When the high noise level events re-occur, the user is not protected. The exposure assessment, and training on the use of the device, should take these factors into account.

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6.3.2.4 Exposure to intermittent or repeated short-term noise

For intermittent or repeated short-term predictable noise exposure, earmuffs and pre-shaped banded earplugs can be an option because they are quick and easy to don and doff. Alternatively, earplugs with a switching function can be used. This allows the users to temporarily select a lower or higher attenuation without the need to don and doff the earplugs.

Level-dependent hearing protectors can also be preferred since they provide protection during the noise exposure, and also allow communication and situational awareness during the quieter periods. Level-dependent hearing protectors can be the preferred option in environments where the user is exposed to sudden and unexpected loud noise.

6.3.2.5 Exposure to impulsive noise

For impulsive noise, the selection is determined by the daily noise exposure level ($L_{\mathrm{EX,8h}}$) and by the highest peak sound pressure level ($L_{p,\mathrm{Cpeak}}$). Level-dependent hearing protectors can offer the optimum solution. For high peak levels, the passive attenuation should be considered when assessing the actual level of protection (see Annex B).

6.3.2.6 Dominant low frequency noise

In environments that are dominated by high-level low frequency noise, hearing protectors with high *L*-values are recommended. Active noise reduction devices can provide additional attenuation in the low frequency range compared to passive devices of the same type. The incorporated electronic sound cancelling system is designed to achieve additional attenuation beyond that of a passive device.

6.3.2.7 High frequency tonal noise

High frequency tonal noise can constitute a higher risk of hearing damage at the same level of exposure compared to a broad band noise. Hearing protectors with high *H*-values are recommended. Devices with a non-flat attenuation curve can be suitable, as long as communication situations are not affected.

6.3.2.8 Informative sounds

6.3.2.8.1 General

In many workplaces it is important that sounds are heard clearly and distinctly. Such sounds can be speech, warning signals, machinery sound or music. If informative sounds of the working process are important, the aspects in 6.3.2.8.2 to 6.3.2.8.6 should be considered.

Dual hearing protection can impair recognition of informative sounds more than single hearing protection.

6.3.2.8.2 Warning signals

Where the recognition of sounds such as warning signals is critical, passive hearing protectors with a flat frequency response can be preferred. In case of non-continuous noise, level-dependent hearing protectors should be considered. The use of hearing protectors provided with entertainment audio is not suitable where there are risks associated with inaudibility of sounds that need to be heard, in order to reduce the risks of accidents, for example from moving vehicles.

Depending on the frequency content of the warning signal, hearing protectors with a suitable frequency response are to be considered. In any case, warning signals need to be clearly recognizable. In case of doubt, a listening check according to EN ISO 7731 should be carried out.

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6.3.2.8.3 Localization of noise sources

Identification of the direction of a sound source is sometimes necessary and can be impaired when hearing protectors are used. Earplugs, some level dependent hearing protectors and flat attenuation hearing protectors can be considered.

6.3.2.8.4 Speech communication

Generally, speech intelligibility through passive hearing protectors is improved with a flat frequency response. Speech intelligibility can also be improved with level-dependent hearing protectors (e.g. for intermittent noise) or hearing protectors with communication facilities.

6.3.2.8.5 Machinery sound

Machinery sounds can contain work related information. Depending on the frequency content of the machinery sound, hearing protectors with a suitable frequency response or level-dependent hearing protectors can be considered.

6.3.2.8.6 Live Music

For music performers and listeners, an attenuation as flat as possible (*H*, *M* and *L*-values being nearly the same) is desirable. This will give the least distortion of the original sound. The attenuation should also be carefully selected. Excessive attenuation should be avoided to encourage uninterrupted use of hearing protectors, see 6.4.2.

6.3.3 Environmental factors (non-acoustic)

6.3.3.1 General

In addition to acoustical factors affecting selection, other environmental factors should also be considered. Table 3 gives general guidance for selecting the appropriate hearing protector when considering environmental factors. Risk assessment can override the effect of the environmental factors on hearing protection selection described in this subclause.

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Table 3 — General guidance for selection of passive hearing protectors in relation to environmental factors

Working environment and activity	Type of hearing protector					
	Earmuffs	Reusable earplugs	Disposable earplugs	Banded earplugs	Custom moulded earplugs	Corded earplugs
Very high temperature or humidity	- [a]	+	+	+	+	+
High airborne particle exposure [b]	o	-	+	-	o	-
Repeated short-term noise exposure	+	o	-	+	-	-
Localization of the noise source	-	+	+	+	+	+
Body movement	o	+	+	o	+	o
Contaminants on the hands such as dirt, biological agents or metal filings	+	o	+[c]	o	o	o
Moving machinery part	+	+	+	o	o	-

Key

- generally not preferable
+ generally preferable
o case specific and subject to risk assessment

[a] Appropriate with absorbent hygiene cushion covers.

[b] Dust can accumulate on the hearing protector and – depending on the type of dust exposure – can irritate the skin. (Typical tasks with high dust exposure are: grinding in containers, fettling, natural stone working.)

[c] Earplugs without grip (especially formable earplugs) should be inserted only with clean hands.

6.3.3.2 Temperature (high and low temperatures and/or humidity)

Physical work, especially at high ambient temperatures and/or humidity, can cause perspiration under earmuff cushions. In such conditions, earplugs can be preferred. If earmuffs are used, thin absorbent hygiene cushion covers can help reduce moisture or sweat build-up. However, it should be noted that these covers can reduce the attenuation of the hearing protector. Products with published attenuation data for a combination of earmuff and hygiene cover should be preferred. Extreme low temperatures can affect the attenuation of hearing protectors because of reduced pliability of the materials.

6.3.3.3 Contamination and unclean working conditions

In unclean working conditions, care should be taken to ensure that the selected hearing protector can be fitted and used without risk of irritation to the user. Contaminants such as dirt, dust, biological agents or metal filings can lead to skin irritation or infection. Dust on the surface of an earmuff cushion or dirt on an earplug can be transferred to the skin causing an adverse skin reaction around the ear or inside the ear canal, for example. Before fitting the hearing protector, the user should ensure that their fitting environment and hands are clean.

All earplugs should be inserted before entering a dusty environment.

6.3.3.4 Moving machinery parts

If working in an environment where machinery has moving parts, corded earplugs, where the cord could be caught by the machine, should be avoided.

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6.3.3.5 Type of work being undertaken

Earmuffs with entertainment audio input can be suitable for users with monotonous or repetitive tasks or when the risk assessment allows it. When selecting such a device the additional noise source represented by the audio input should be taken into account (see 6.2.3.6 and Annex D). Warning signals need to be clearly recognizable. In case of doubt, carry out a listening check in accordance with EN ISO 7731.

6.3.3.6 Explosive atmospheres

Where hearing protectors are selected for use in potentially explosive atmospheres, care should be taken to ensure that they fulfil requirements for intrinsic safety in the particular environment.

6.3.3.7 Body movement

Rapid body movements, movements induced by machine or vehicle vibrations, or rubbing of protective clothing can affect the fit and comfort of the hearing protector.

6.4 Communication

6.4.1 Essential work or safety-related speech communication, especially speech intelligibility

Where the recognition of sounds such as speech is critical, careful selection of attenuation is necessary to ensure a good balance between speech intelligibility and protection (with due consideration to the risk of excessive attenuation). Passive hearing protectors with a flat frequency response are an option. Alternatively, hearing protectors with additional (electronic) sound-restoration or active noise reduction can be considered.

Where communication by audio input is required, hearing protectors are available with one-way or two-way communication functionality. This can be achieved via wired or wireless audio input or integrated into the device.

For essential safety-related communication, these hearing protection devices can offer higher sound pressure levels effective to the ear to achieve a good signal to noise ratio. This can be potentially damaging, if the user is exposed to these levels for prolonged periods. See Annex D for a calculation method for hearing protectors with audio input. Methods for assessing the speech intelligibility can be found in EN ISO 9921.

6.4.2 Effects of excessive attenuation

Caution should be exercised to ensure that the user is not provided with a hearing protector that gives unnecessarily high attenuation. Such passive hearing protectors can cause difficulties with communication and audibility of warning signals. Users can experience discomfort or feel isolated from the environment. As a result, hearing protectors are often not used for the entire exposure time which can lead to risk of hearing damage.

6.5 Compatibility with other Personal Protective Equipment (PPE)

In the selection of hearing protectors, it is important to ensure that the hearing protector's performance is not impaired by the use of other PPE. For more information see 7.4.

6.6 Selection according to hearing protector type and situation of use

6.6.1 Earmuffs – General

Earmuffs can be preferred:

a) in cases where the hearing protector is frequently donned and doffed due to repeated short term noise exposure (banded earplugs are also suitable as an alternative);

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b) in cases where earplugs result in discomfort;

c) whenever earplugs cause inflammation, irritation or other adverse effects in the ear canal.

Earmuffs of low mass are preferred, if sound attenuation requirements are met.

6.6.2 Earmuffs attached to head protection and/or face protection devices

Earmuffs attached to head and/or face protection devices are intended for use when a user requires simultaneous head and/or face and hearing protection. Only use combinations of protective equipment that have been appropriately tested and certified. Information about approved combinations is contained in the user information from earmuff manufacturers.

6.6.3 Earplugs – General

Earplugs can be preferred:

a) where moisture or sweat build-up can occur under earmuffs;

b) to help minimize potential interference if multiple items of PPE are used at the same time, e.g. respirators, goggles or face shields.

In general, hearing protectors tend to impair the localization of noise sources, particularly when the attenuation is high. Preference can be given to earplugs if identification of the sound direction is required.

Banded or corded earplugs can be preferred in cases where they are frequently inserted and removed. Such devices also lower the risk of loss of one or both earplugs. If working in an environment where machinery has moving parts, corded earplugs should be avoided.

Custom moulded earplugs, properly fitted to the ear canal, can be preferred for long duration use because they are less likely to work loose during the period of use, are easy to insert and can provide a repeatable fit.

6.6.4 Hearing protectors with additional function modes

6.6.4.1 Earmuffs and earplugs with level-dependent attenuation can be preferred if the working noise environment contains intermittent or impulsive noise. These products can provide situational awareness and good speech intelligibility when the noise levels are low. For high noise levels, the quality of the transmission depends on the product.

6.6.4.2 Communication earmuffs and earplugs can be preferred if work activities require the ability to communicate using a radio, a mobile phone or other communications system.

6.6.4.3 Active noise reduction earmuffs are suitable for low-frequency noise with a high sound pressure level.

6.6.4.4 Earmuffs or earplugs with entertainment audio input can be suitable for users with monotonous or repetitive tasks, where situational awareness or perception of informative sounds of the working process is not required. However, warning signals need to be clearly recognizable at any time.

6.7 Medical factors

Before any type of hearing protector is selected, medical factors should be considered. These can include but are not limited to irritation of the ear canal, earache, discharge, infections, tinnitus, hearing impairment, previous ear surgery, or any ongoing treatment for ear disease or skin conditions.

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Individuals with any such medical problems should be referred for specialist advice as to the most suitable type of hearing protector.

Individuals with an existing hearing impairment can find that the use of hearing protectors leads to additional difficulties hearing and localizing important sounds, such as fire alarms, warning signals, machinery sounds and communications. It is important to note that emergency signals should not be limited to audible signals, and care should be taken to ensure these alerts are accessible to all individuals. In general, a hearing protector with a flat-frequency response could improve the situation compared to conventional passive hearing protectors. Sound restoration hearing protectors can also improve situational awareness and communication if using a hearing aid is not possible or permitted, or when used in combination with a hearing aid. The combination of earmuffs and a hearing aid requires individual assessment to avoid exceeding the national regulation level.

6.8 Ergonomics and fitting

Factors to be considered include, but are not limited to, size and shape of the head, ear canal and pinna.

Comfort is an important element in the selection process of a hearing protector. If the chosen product is not comfortable, it is unlikely to be used. Wherever possible, the user should be allowed to make a personal choice of a hearing protector from among adequate and suitable types.

For earmuffs, the comfort of a hearing protector can be affected by the size, mass, cushion pressure, headband force, adjustability, type of material used, the design and construction of the device.

For earplugs, the comfort can be affected by the size, pressure, ease of fit and removal, the material, design and construction of the device. If a user has small ear canals or sensitive ear canal tissue, the use of earplugs could be inappropriate. In such cases earmuffs can be the preferred option.

At low ambient temperature, foam earplugs can feel too hard but once they warmed-up to body temperature prior to insertion they become soft and pliable again. For earmuffs temperature can affect the characteristics of the earmuff cushions.

7 Use

7.1 General

Effective explanation of the need to use hearing protection can help to overcome initial resistance. A sense of isolation can be experienced when hearing protectors are used for the first time.

When a hearing protector is used, the low frequency energy transmitted by the ear canal wall (bone conduction path) will be more effectively radiated and coupled to the eardrum. This phenomenon is known as the occlusion effect. The user can experience their own voice differently and a sense of pressure or blockage in the ear.

Further guidance on improving field performance and special instructions for use is available in Annex E.

7.2 Availability of hearing protectors

Personnel should not enter noise hazard areas without using appropriate hearing protectors. Where disposable hearing protectors are used, they should be available in sizes that will fit the intended users at all points of entry to the designated noise hazard area. For visitors suitable hearing protectors should be available and strictly used.

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7.3 Correct fitting

7.3.1 General

Head and ear canal shapes vary considerably from one person to another. To achieve a good fit, it is therefore essential that any hearing protector is correctly sized or adjusted for each individual. In this regard, it should be noted that hearing protectors are available in various sizes. Selection of the correct size should be based on achieving a good seal and then comfort.

NOTE For comfort see 6.8.

In order to achieve the desired attenuation, hearing protectors should be fitted in accordance with the manufacturer's instructions and fit tested where possible.

Fit testing is beneficial for all types of hearing protectors to maximize effectiveness because there can be differences between the laboratory sound attenuation and the attenuation obtained in the field. For earplugs this difference can be higher than for earmuffs. The result of fit testing is often expressed as a personal attenuation rating (PAR).

For custom moulded earplugs poor fit can be due to:

— errors in imprint taking;

— errors in manufacturing;

— poor insertion;

— lack of training.

For universal earplugs poor fit is generally due to:

— poor insertion, which can be caused by incorrect size selection (too small or too large);

— an incompatibility between the shape of the earplug and that of the ear canal;

— a lack of training.

Further guidance on fit testing methods can be found in EN 17479. See Annex F for additional information on the applicability of different fit testing methods.

In cases where hearing protectors have one cup or plug intended only for the left ear and another for the right ear, care should be taken to ensure that the hearing protectors are used in the correct orientation.

7.3.2 Earmuffs

Earmuffs should be used with the head band or neck band correctly positioned and adjusted.

For mounted earmuffs, a suitable combination of carrier and earmuffs is important to ensure appropriate head band force, cushion pressure and attenuation when the earmuff is in the working position. Only tested and certified combinations should be used, further information is available from the device manufacturers.

Earmuffs are generally easy to fit, but large or small head sizes can require specific selection or adjustments. This particularly applies to mounted earmuffs.

Ideally, confirmation of the right choice of the earmuff, including verification that the size and fit are appropriate for each individual, will be achieved using a fit testing method. The different methods available are described in EN 17479:2021, Clause 4. The recommended method for earmuffs is the

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MIRE method described in EN 17479:2021, 4.4.1.3. The uncertainty associated with the chosen method should be carefully evaluated and the results interpreted in the context of measurement of uncertainty (see EN 17479:2021, Clause 8).

7.3.3 Earplugs

Earplugs should be correctly inserted into the ear canal, providing a good seal. Failure to do so can lead to poor sound attenuation, particularly at low frequencies.

In contrast to earmuffs the fitting of earplugs requires more attention. A combination of training and a variety of products and/or sizes should be offered for fitting earplugs.

Ideally, confirmation of the right choice of earplugs, including verification of the correct size and fit appropriate for each individual, can be achieved using a fit testing method. The different methods available are described in EN 17479:2021, Clause 4. The most appropriate method will be selected in accordance with EN 17479:2021, Clause 6. It will depend on the earplug selected, the main objectives, the time available per individual and the environment in which the tests can be performed. See EN 17479:2021, Clause 5, which describes the test protocols and their requirements. The uncertainty associated with the chosen method should be carefully evaluated and the results interpreted in the context of this uncertainty (see EN 17479:2021, Clause 8).

When in use, earplugs can slowly become loose due to jaw movement, for example. Check and refit as appropriate.

The parts of earplugs that come into direct contact with the ear canal should be clean and earplugs should always be inserted with clean hands and fingers.

7.4 Simultaneous use of hearing protectors with other PPE

7.4.1 General

It is not uncommon for personnel working in a noisy environment to wear other types of PPE in addition to hearing protection. This can lead to a potential reduction in hearing protector attenuation. Check with the manufacturer for any available data for PPE combinations. The subclauses that follow concern combinations of PPE that are not covered by EN 352-1 to EN 352-3.

When a hearing protector is used in combination with other PPE equipped with acoustic warning signals, care should be taken to ensure that the user's recognition of these warning signals is not impaired.

NOTE For other effects not related to the sound attenuation that can be expected for combinations of PPE see EN 17558.

7.4.2 Protective clothing

Protective clothing should be used over any hearing protector. Using hearing protectors over clothing is likely to greatly reduce their attenuation.

7.4.3 Spectacles

The side arms of spectacles should be of a low-profile type so as not to disturb the acoustic seal of the earmuff against the head. In order to minimize the acoustic leakage, avoid using spectacles with broad side arms. Earplugs or earmuffs with wide and pliable cushions may be preferred.

7.4.4 Goggles

Care should be taken to ensure that the head straps do not interfere with the earmuff cushions and seals.

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7.4.5 Face shields

The design of the face shield should be such as to prevent it interfering with the hearing protector when in use.

7.4.6 Hoods

Hearing protectors are usually used under a hood. Some hoods have been designed to be used under specific hearing protectors. Further advice should be available from manufacturers of such devices.

7.4.7 Head protection

As an alternative to mounted earmuffs that require a carrier, e.g. safety helmet, some earmuffs are designed to be used independently with head protection. These can be held in place with a neck band or a chin band and a head strap, if applicable. Care should be taken to ensure that the head protection does not interfere with the fitting of the earmuff.

When using earplugs underneath head protection that covers the ears, select earplugs that have minimal protrusion outside of the ear canal to prevent mechanical coupling of the head protection to the earplugs and to optimize comfort.

7.4.8 Respiratory protection devices

Care should be taken to ensure that the retaining harness or head strap of the respirator does not interfere with the seal between the earmuff and the side of the head.

7.5 Speech intelligibility and signal audibility

It is a natural reaction to drop the voice level when hearing protectors are used. However, it is important that users maintain or even increase their voice level to improve communication. Warning signals, alarms or call signals in the noisy area should be carefully selected so that they are audible to those who use hearing protection (see EN ISO 7731 for guidance). The audibility of any desired signal should be maintained by testing it under actual working conditions, allowing for variations in sound pressure levels (e.g. time of day/work process).

If the sound pressure levels are loud enough to interfere with the audibility of such signals it can be necessary to adopt an additional backup visual alarm system. Hearing protectors with communication facilities can be preferable if speech intelligibility or signal audibility is critical.

Poor speech intelligibility can prompt the user to temporarily remove the hearing protector for the purposes of communication. This is likely to lead to a dramatic reduction in effective attenuation, see 7.7.

7.6 Instruction and training

7.6.1 General

All personnel who need to use hearing protectors should be provided with sufficient information and adequate training in their correct use.

7.6.2 Information

In particular, information should be provided to the user on the following points:

a) risk of hearing damage if hearing protectors are not used;

b) damaging sound pressure levels;

c) influence of the period of use on the protection achieved;

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d) availability of hearing protectors;

e) importance of the fit of hearing protectors on performance and the protection achieved;

f) audibility of speech and/or warning and alerting sounds;

g) manufacturer's instructions for use.

First time use requires a period of time for the user to adapt and familiarize with the new acoustical situation. This applies, in particular, to musicians.

Where required, additional information should be provided for the following:

1) use of hearing protectors in particular orientations;

2) compatibility of other PPE with earmuffs and banded earplugs;

3) changes of properties of hearing protectors over time as well as necessary inspection and maintenance.

7.6.3 Special instructions for use including training

This guidance can be used to minimize the difference between the sound attenuation according to the EU type-examination and the sound attenuation in practise.

For details see Annex E.

NOTE Hearing protection being in the highest risk category in the PPE Regulation needs careful attention in selecting, using and maintaining personal protective equipment. In some countries, there are specific requirements for the use of hearing protection, namely instructions with practical training.

7.7 Period of use

To be effective, hearing protectors need to be used all the time in environments where harmful sound pressure levels exist. If hearing protectors are removed, even for short periods of time, the effective attenuation and protection is greatly reduced. Figure 1 illustrates the effect of removing the hearing protector for a period of time during an 8 h exposure. When people do not use hearing protectors for the complete noise exposure time, the limiting factor, in terms of attenuation, will become the time spent without the hearing protector rather than the performance of the protector. The loss in attenuation is more significant for higher attenuating devices, even when they are removed for a relatively short period of time during the exposure.

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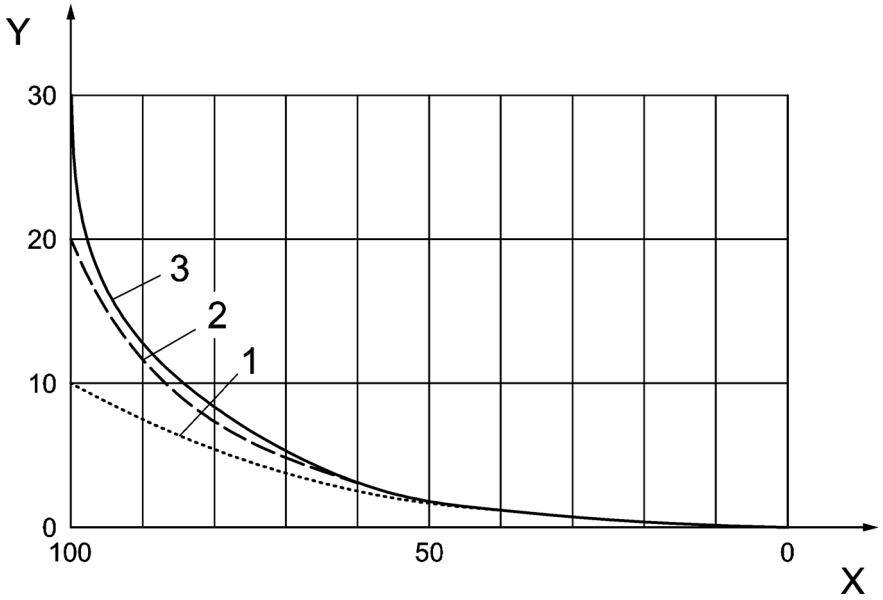

Key

x-axis: exposure time with hearing protector in %

y-axis: effective attenuation in dB

1 hearing protector with an attenuation of 10 dB

2 hearing protector with an attenuation of 20 dB

3 hearing protector with an attenuation of 30 dB

Figure 1 — Effective protection provided by hearing protectors with decreased usage time over an 8 h shift

EXAMPLE If used for 4 h of an 8 h exposure period (i.e. 50 % of the time), the effective protection provided by any hearing protector is not more than 3 dB.

7.8 Leisure activities

The use of hearing protectors should be encouraged when exposure to noise during leisure activities is hazardous.

Risk assessors should consider potential hazardous noise exposure from leisure activities together with occupational noise exposure when assessing the total noise exposure.

8 Care and maintenance

8.1 General

Hearing protectors should be regularly maintained and cleaned in order to avoid loss of attenuation. The same earplugs should never be used by more than one person. In circumstances where the same earmuffs are used by more than one person (e.g. earmuffs supplied for use by visitors) they should be hygienically cleaned between uses. The use of disposable earmuff cushion covers can help improve hygiene, however, it should be noted that this can result in loss of attenuation. Information on the care and maintenance of hearing protectors should be given to all persons who use hearing protectors and should be repeated at regular intervals.

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8.2 Hygiene and cleaning

Contamination of hearing protectors by foreign materials, chemical solvents, liquid residues, dusts, particulate matter, etc. could cause skin irritation or abrasions. The user should ensure that their hands are clean when handling hearing protectors, especially earplugs. Medical assistance should be sought if skin irritation occurs during or following the use of hearing protectors. The use of disposable earmuff cushion covers can improve hygiene, however, it should be noted that this can result in a loss of attenuation.

Earmuffs (especially the cushions) and reusable earplugs should be cleaned and stored following the manufacturer's instructions. Care should be exercised when cleaning hearing protectors that incorporate electronic or other specialized components and they should not be immersed in fluid.

8.3 Inspection and replacement

Hearing protectors should be replaced in accordance with the manufacturer’s instructions.

Additionally, hearing protectors should be inspected at frequent intervals for damage due to mechanical or electrical failure, ageing, accident or abuse. Signs can include evidence of hardness, brittleness or cracking, for example. Head bands should not be subject to intentional deformation.

Replaceable parts such as earmuff cushions and inserts, and earplug ear tips, should be replaced in accordance with the manufacturer’s instructions.

8.4 Storage

Hearing protectors should be stored in a clean, dry and uncontaminated environment, in accordance with the manufacturer’s instructions.

8.5 Disposal

Disposal arrangements for used hearing protectors should follow manufacturer’s instructions. Additionally, national regulations can apply. Such disposal arrangements should ensure that the hearing protectors cannot be inadvertently reused and that they do not cause environmental damage.

9 Sustainability

When selecting and using hearing protectors, aspects of sustainability amongst other product attributes, should also be considered.

Life cycle assessment or environmental product declarations are tools to assess the impact of a product on the environment.

Examples of specific approaches to address sustainability can be found in Annex H.

The primary function of a hearing protector is to reduce hazardous noise for the user. This shall always remain the utmost priority and should never be compromised. To help protect the user’s hearing, the hearing protector shall fit properly and be worn at all times when exposed to hazardous noise. As such, product attributes which help meet compliance should remain high priority in the selection criteria process. These attributes include, but are not limited to, the model type, comfort, ease of use, fit, compatibility with other PPE and the situations of use.

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Annex A
(informative)

Methods for estimating the level effective to the ear when selecting a hearing protector based on the national regulation level L'_{NR}

A.1 General

This annex describes methods for estimating the level effective to the ear when a hearing protector is used in continuous, fluctuating and intermittent noise and gives examples. The attenuation of most hearing protectors varies with frequency. Therefore, to correctly estimate the sound pressure level effective to the ear when a hearing protector is used, the frequency characteristics of both the noise environment and the protector should be taken into account. This annex describes four different estimation methods to do this.

When the sound pressure level effective to the ear has been estimated, the daily noise exposure level effective to the ear shall be calculated, allowing for the combination of the noise environment and taking into account the duration of exposure. The method for predicting the hearing protector attenuation should be chosen according to the information available for the workplace noise as shown in Table 2.

All methods are based on sound attenuation data measured according to EN ISO 4869-1. The methods are also applicable to the total (passive and active) attenuation data as specified in EN 352-5 for ANR earmuffs. In addition, possible compensation for expected effects in practice can be made (see 6.2.2 and national guidelines). Used in this manner, the main purpose of the selection methods is to identify whether a protector is likely to be suitable for the user(s), subject to a given noise exposure.

The octave band method (A.2) is the most accurate method. It involves both the octave band sound pressure levels at the workplace and the octave band attenuation data of the hearing protector, i.e. assumed protection value (APV_f), being assessed (APV_f is mean attenuation minus one standard deviation for octave band centre frequency f). Without the use of a computer or calculator program this method is complex and time consuming.

The *HML* method (A.3) specifies three attenuation values, *H*, *M* and *L*, calculated from the individual attenuation values of the hearing protector for high (*H*), middle (*M*) and for low (*L*) frequency noise. These values, together with the measured A-weighted and C-weighted sound pressure levels, are used to calculate the predicted noise level reduction (*PNR*). The *PNR* is then subtracted from the observed A-weighted sound pressure level to calculate the A-weighted sound pressure level effective to the ear when the hearing protection is used.

The *HML* check method (A.4) is an abbreviation of the *HML* method, that uses examples of noise sources which typically have a certain frequency content.

The *SNR* method (A.5) uses a single attenuation value, the Single Number Rating (*SNR*). The *SNR*-value is subtracted from a measured C-weighted sound pressure level, to calculate the A-weighted sound pressure level effective to the ear when the hearing protector is used.

The choice of method depends on the data available. In general, octave band sound pressure levels are often not available. This together with the added calculation complexity, means that the octave band method is not often used. It is therefore suggested to consider the *HML* method.

However, the octave band method is clearly recommended if:

— the noise is dominated by a significant high or low frequency content;

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— the noise has a pure tone character; and/or

— the attenuation of the protector is notably frequency dependent (which is often the case with low attenuation devices).

It is recommended to use the APV_f, H, M, L or SNR with a protection performance of 84 % (equals mean value minus one standard deviation). This value is also applied in EN 352-1, EN 352-2 and EN 352-3.

The reason for using mean values and standard deviations is that the attenuation of any hearing protector will show variation between users due to person-related physical factors as well as variation in fitting. This variation is expressed by the standard deviation value. The mean and standard deviation values are derived from the data obtained during the laboratory test for the type approval. With a protection performance of 84 % there is statistically an 84 % chance that the value is reached or exceeded, or a 16 % chance that the value is below the presented value.

NOTE EN ISO 4869-2 allows the calculation of the attenuation with other values of protection performance. For example, subtracting two standard deviations gives a protection performance of 98 %.

All four methods (A.2 to A.5) predict the overall A-weighted sound pressure level effective to the ear when the hearing protector is used ($L'_{p,\mathrm{A}}$). For changing noise environments and/or exposure times, the daily noise exposure level should be calculated and used. In some of these cases it can be appropriate, for comfort reasons, to use more than one hearing protector, each one selected according to the $L_{p,\mathrm{A,eq}}$ for the different noise environments.

A full data label describing the attenuation of a hearing protector is shown in Table A.1 and Table A.2. Table A.1 contains octave band data from 63 Hz to 8 kHz and Table A.2 contains H, M, L and SNR data.

Table A.1 —Octave band attenuation data for a hearing protector

Octave band centre frequency in Hz	63	125	250	500	1k	2k	4k	8k
Mean value in dB	7,4	9,5	14,2	18,8	21,7	29,6	34,9	32,4
Standard deviation in dB	2,4	2,5	2,8	3,1	2,3	5,2	2,3	2,7
Assumed protection value (APV_f) in dB	5,0	7,0	11,4	15,7	19,4	24,4	32,6	29,7

Table A.2 — *H*, *M*, *L* and *SNR* data for a hearing protector

	H	***M***	***L***	***SNR***
Mean value in dB (H_m, M_m, L_m, SNR_m)	28,2	21,1	15,9	24,0
Standard deviation in dB (H_s, M_s, L_s, SNR_s)	3,0	2,4	2,6	2,5
Mean minus standard deviation in dB (rounded to the nearest integer) (H, M, L, SNR)	25	19	13	22

Octave band data

Table A.1 shows the octave band data. The data for each frequency band are:

— the mean attenuation in decibels;

— the standard deviation in decibels;

— the “Assumed Protection Value” (APV_f). This is the difference between mean and standard deviation.

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HML data

The attenuation of most protectors is frequency dependent. *H-M-L* is a three number data set specifying the typical attenuation that can be achieved by a protector for noises that are respectively predominantly "High", "Medium" or "Low" in frequency content. The *H*, *M* and *L* values correspond to $L_{p,C} - L_{p,A}$ differences of −2 dB, +2 dB and +10 dB respectively.

The difference between C-weighted and A-weighted sound pressure levels is used to determine which formulae and value to use. The higher the difference, the more low-frequency the noise is.

H, *M* and *L* represent assumed protection with a protection performance of 84 % (equals mean attenuation minus one standard deviation). H_m, H_s, M_m, M_s, L_m and L_s show the mean attenuation and standard deviations of the corresponding *H*, *M* and *L*-values.

Table A.2 shows *H*, *M* and *L* data. The data for each frequency character are:

— the mean attenuation (H_m, M_m, L_m) in decibels;

— the standard deviation (H_s, M_s, L_s) in decibels;

— the mean attenuation minus one standard deviation (*H*, *M* and *L* with a protection performance of 84 %) in decibels.

SNR data

The *Single Number Rating* value is based on the C-weighted sound pressure level outside the protector to predict the effective A-weighted sound pressure level to the ear. The higher the frequency content, the more accurate the result. The lower the frequency content the more chance for incorrect estimation.

SNR represents assumed protection with a protection performance of 84 % (equals mean attenuation minus one standard deviation). SNR_m and SNR_s show the mean attenuation and standard deviation of the corresponding *SNR*-value.

Table A.2 shows *SNR* data. The data are:

— the mean attenuation (SNR_m) in decibels;

— the standard deviation (SNR_s) in decibels;

— the mean attenuation minus one standard deviation (*SNR* with a protection performance of 84 %) in decibels.

Calculation examples are provided in the following subclauses A.2 to A.5.

For the examples given in A.2 to A.5, the data from Table A.1, Table A.2 and Table A.3 have been applied.

Table A.3 — Continuous noise environment throughout the day

Octave band centre frequency in Hz	63	125	250	500	1k	2k	4k	8k
Octave band level ($L_{p,oct}$) in dB:	80	84	86	88	97	99	97	96

$L_{p,A}$ = 104 dB

$L_{p,C}$ = 103 dB

$L_{p,C} - L_{p,A}$ = −1 dB

Noise Class *HM*

As a basis for assessment the following Table A.4, referring to a national regulation level, can be used.

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Table A.4 — Example of assessment on the basis of a national regulation system

Level effective to the ear ($L'_{p,\mathrm{A,eq}}$ in dB)	Protection rating
$L'_{\mathrm{NR}} < L'_{p,\mathrm{A,eq}}$	Insufficient
$L'_{\mathrm{NR}} - 5 < L'_{p,\mathrm{A,eq}} \leq L'_{\mathrm{NR}}$	Acceptable
$L'_{\mathrm{NR}} - 10 \leq L'_{p,\mathrm{A,eq}} \leq L'_{\mathrm{NR}} - 5$	Good
$L'_{\mathrm{NR}} - 15 \leq L'_{p,\mathrm{A,eq}} < L'_{\mathrm{NR}} - 10$	Acceptable
$L'_{p,\mathrm{A,eq}} < L'_{\mathrm{NR}} - 15$	Risk of excessive attenuation[a]

[a] Sound attenuation can be too high and speech intelligibility can be hindered, check for communication and acoustical isolation.

For assessing the risk to hearing from long term exposure the daily noise exposure level $L'_{\mathrm{EX,8h}}$ shall be used instead of $L'_{p,\mathrm{A,eq}}$.

A.2 Octave band method

The octave band method is a noise reduction calculation involving the workplace octave band sound pressure levels and the octave band attenuation data (see A.1) for the hearing protector being assessed.

Step 1: Calculate the A-weighted sound pressure level ($L'_{p,\mathrm{A}}$) under the hearing protector using the following formula:

$$L'_{p,\mathrm{A}} = 10 \lg \sum_{f=125}^{8\,000} 10^{0,1\left(L_{p,f} + A_f - APV_f\right)} \tag{A.1}$$

where

f represents the centre frequency of the octave band in Hz;

$L_{p,f}$ is the octave band sound pressure level ($L_{p,\mathrm{oct}}$) of the noise in dB in octave band f;

A_f is the frequency weighting A in dB for octave band centre frequency f;

APV_f is the assumed protection value of the hearing protector in dB for octave band centre frequency f.

NOTE For a definition of A_f, see EN 61672-1.

If the attenuation data of 63 Hz is available the calculation can commence at this frequency.

Step 2: Round to the nearest integer.

Table A.5 contains a calculation example.

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Table A.5 — Example of calculation of $L'_{p,A}$ using the octave band method (using data given in Table A.1 and Table A.3)

Octave band centre frequency in Hz	63	125	250	500	1k	2k	4k	8k
Measured octave band sound pressure levels of the noise in dB (Row 1)	80	84	86	88	97	99	97	96
A-weighting in dB (Row 2)	−26,2	−16,1	−8,6	−3,2	0	+1,2	+1,0	−1,1
Add Row 2 to Row 1 (Row 3)	53,8	67,9	77,4	84,8	97,0	100,2	98,0	94,9
Assumed protection values of the hearing protector (Row 4)	5,0	7,0	11,4	15,7	19,4	24,4	32,6	29,7
Subtract Row 4 from Row 3 and multiply by 0,1 (Row 5)	4,88	6,09	6,60	6,91	7,76	7,58	6,54	6,52
NOTE Modern instrumentation can combine rows 1 to 3 and directly display the A-weighted spectrum.								

Step 1: Calculate $L'_{p,A}$ as follows:

$$L'_{p,A} = 10\lg\left(10^{4,88} + 10^{6,09} + 10^{6,60} + 10^{6,91} + 10^{7,76} + 10^{7,58} + 10^{6,54} + 10^{6,52}\right) \quad \text{(A.2)}$$

$$L'_{p,A} = 80,6\ \text{dB}(\text{A}) \quad \text{(A.3)}$$

Step 2: Round this figure to the nearest integer; the A-weighted sound pressure level under the hearing protector is 81 dB(A).

Assessment: For a value of L'_{NR} = 85 dB(A), the $L'_{p,A}$ of 81 dB(A) lies between 0 and −5 dB of the L'_{NR}. The choice of this hearing protector is therefore considered “acceptable” for the given noise (see Table A.4).

A.3 *HML* method

The *HML* method is based on three attenuation values, *H*, *M* and *L* (see A.1). These values, when combined with a measurement of the A-weighted and C-weighted sound pressure levels of the noise, are used to calculate the predicted noise level reduction (*PNR*). The *PNR* is then subtracted from the observed A-weighted sound pressure level to calculate the A-weighted sound pressure level effective to the ear when the hearing protector is used ($L'_{p,A}$).

Step 1: Calculate the difference between the C-weighted and the A-weighted sound pressure levels of the noise ($L_{p,C} - L_{p,A}$).

Step 2: Calculate the predicted noise level reduction (*PNR*) according to one of the following formulae:

Either:

$$PNR = M - \frac{H - M}{4}\left(L_{p,C} - L_{p,A} - 2\right)\text{dB};\ \text{for}\left(L_{p,C} - L_{p,A}\right) \le 2\ \text{dB} \quad \text{(A.4)}$$

Or:

$$PNR = M - \frac{M - L}{8}\left(L_{p,C} - L_{p,A} - 2\right)\text{dB};\ \text{for}\left(L_{p,C} - L_{p,A}\right) > 2\ \text{dB} \quad \text{(A.5)}$$

Step 3: Calculate the A-weighted sound pressure level effective to the ear according to the following formula:

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$$L'_{p,\mathrm{A}} = L_{p,\mathrm{A}} - PNR \tag{A.6}$$

Step 4: Round $L'_{p,\mathrm{A}}$ to the nearest integer.

As an example for assessment compare $L'_{p,\mathrm{A}}$ to Table A.4.

Example of calculation of $L'_{p,\mathrm{A}}$ using the HML method (using data given in Table A.2)

Step 1: The difference between the C-weighted and the A-weighted sound pressure levels of the noise is −1 dB.

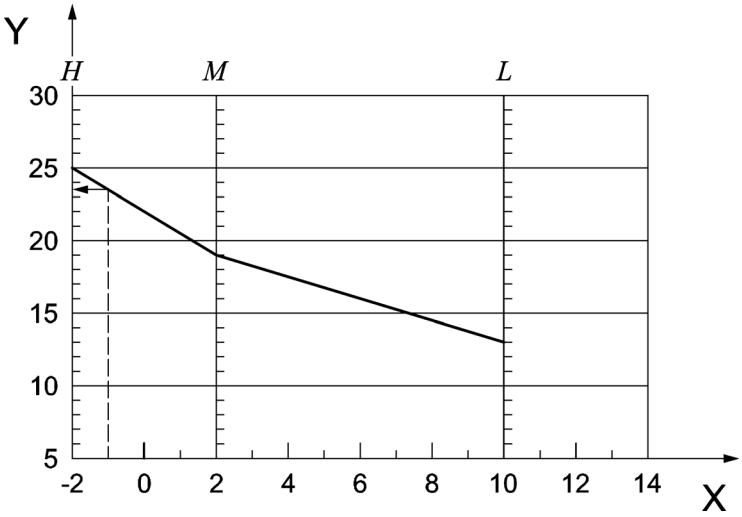
Step 2: Instead of calculating the *PNR* value using one of the two equations, a graph can be used with marked-up *H*, *M* and *L* values as shown in Figure A.1. Using the graph as an example of illustration with ($L_{p,\mathrm{C}} - L_{p,\mathrm{A}}$) = −1 dB, the *PNR* can be read across as 23,5 dB.

Step 3:

$$L'_{p,\mathrm{A}} = 104\ \mathrm{dB} - 23{,}5\ \mathrm{dB} = 80{,}5\ \mathrm{dB}\left(\mathrm{A}\right) \tag{A.7}$$

Step 4: Round $L'_{p,\mathrm{A}}$ to the nearest integer.

Assessment: For a value of L'_{NR} = 85 dB(A), the $L'_{p,\mathrm{A}}$ of 81 dB(A) lies between 0 and −5 dB of the L'_{NR}. The choice of this hearing protector is therefore considered “acceptable” for the given noise (see Table A.4).



Key

x-axis: $L_{p,\mathrm{C}} - L_{p,\mathrm{A}}$ in dB

y-axis: *PNR* in dB

H: high-frequency attenuation value

M: middle-frequency attenuation value

L: low-frequency attenuation value

Figure A.1 — Graph showing how the *PNR* value is read across without calculation

A.4 *HML* check method

The *HML* check method is an abbreviation of the *HML* method. In general, it needs no knowledge of the C-weighted sound pressure level or the level difference ($L_{p,\mathrm{C}} - L_{p,\mathrm{A}}$).

Step 1: Decide, by a listening check of the noise at the workplace and consultation of the list of examples in Tables A.6 and A.7, if the noise is:

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categorized as middle to high frequency noise ($L_{p,C} - L_{p,A} < 5$ dB) (see Table A.6: Examples of noise sources - Noise Class *HM*) – Go to Step 2B

or

categorized as low frequency dominant noise ($L_{p,C} - L_{p,A} \geq 5$ dB) (see Table A.7: Examples of noise sources – Noise Class *L*) – Go to Step 2A

NOTE The lower the value for the difference between $L_{p,C}$ and $L_{p,A}$ the greater the high frequency content of the noise.

Table A.6 —Examples of noise sources — Noise Class *HM* (middle to high frequency noises) ($L_{p,C} - L_{p,A} < 5$ dB)

Flame cutting	High speed web-fed rotary presses
Sugar-coating machines	Circular saw
Compressed air nozzles	Impact tools with high repetition rate impulses
Power nail drivers	Grinding machines
Folding/beading machines	Spinning frames
Bottle filling machines	Hosiery/knitting machines
Trimming of castings	Abrasive cutting machines
Wood working machines	Power weaving looms
Hydraulic pumps	Centrifuges
Honing machines	

Table A.7 —Examples of noise sources – Noise Class *L* (dominant low frequency noises) ($L_{p,C} - L_{p,A} \geq 5$ dB)

Excavators	Compressor units (piston)
Motor-generator sets	Converter units
Electric melting furnaces	Cupola furnaces
Combustion furnaces	Die-casting machines
Annealing furnaces	Earth-moving equipment
Blast furnaces	Blast-cleaning machines
Crushing mills	Large diesel engines (locomotives, ships)

Step 2A: Subtract the ***L*-value** from the A-weighted sound pressure level ($L'_{p,A} = L_{p,A} - L$):

if $L'_{p,A} > L'_{NR}$ the protection is insufficient; try another hearing protector with higher attenuation;

if $L'_{p,A} \leq L'_{NR}$ the sound attenuation of the hearing protector is sufficient;

if $L'_{p,A} \geq L'_{NR} - 15$ dB the sound attenuation is “acceptable” or “good”.

Step 2B: Subtract the ***M*-value** from the A-weighted sound pressure level ($L'_{p,A} = L_{p,A} - M$):

if $L'_{p,A} > L'_{NR}$ go to Step 2C;

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if $L'_{p,A} \leq L'_{NR}$ the sound attenuation of the hearing protector is sufficient;

if $L'_{p,A} \geq L'_{NR} - 15$ dB the sound attenuation is “acceptable” or “good”.

Step 2C: Subtract the ***H*-value** from the A-weighted sound pressure level ($L'_{p,A} = L_{p,A} - H$):

if $L'_{p,A} > L'_{NR}$ the protection is insufficient; try another hearing protector with a higher attenuation;

if $L'_{p,A} \leq L'_{NR}$ the hearing protector can be suitable; obtain additional information for the noise and go to A.2, A.3 or A.5.

Example of calculation of $L'_{p,A}$ using the HML check method (using data given in Table A.2)

Step 1: Given a high frequency noise of Noise Class *HM* with $L_{p,A}$ = 104 dB, go to Step 2B.

Step 2B: $L'_{p,A} = L_{p,A} - 19$ dB = 85 dB(A)

Assessment: For a value of L'_{NR} = 85 dB(A), the $L'_{p,A}$ of 85 dB(A) is equal to the L'_{NR}. The choice of this hearing protector is therefore considered just “acceptable” for the given noise (see Table A.4).

A.5 *SNR* method

The *SNR* method is based on a single attenuation value, the Single Number Rating (*SNR*) (see A.1). Like the *PNR*, the *SNR* is subtracted from an overall sound pressure level measurement, in this case the C-weighted sound pressure level, to calculate the A-weighted sound pressure level effective to the ear when the hearing protector is used.

The predicted A-weighted sound pressure level under the hearing protector ($L'_{p,A}$) can be calculated:

on the basis of the C-weighted sound pressure level at the workplace ($L_{p,C}$) according to

$$L'_{p,A} = L_{p,C} - SNR \tag{A.8}$$

or

on the basis of $L_{p,A}$ and the difference between C-weighted and A-weighted sound pressure levels according to

$$L'_{p,A} = L_{p,A} + \left(L_{p,C} - L_{p,A}\right) - SNR \tag{A.9}$$

where applicable, round $L'_{p,A}$ to the nearest integer.

As an example for assessment compare $L'_{p,A}$ to Table A.4.

Example of calculation of $L'_{p,A}$ using SNR method (using data given in Table A.2)

The C-weighted sound pressure level ($L_{p,C}$) is 103 dB, therefore $L'_{p,A}$ is calculated according to:

$L'_{p,A}$ = 103 dB – 22 dB

$L'_{p,A}$ = 81 dB(A)

Assessment: For a value of L'_{NR} = 85 dB(A), the $L'_{p,A}$ of 81 dB(A) lies between 0 and –5 dB of the L'_{NR}. The choice of this hearing protector is therefore considered “acceptable” for the given noise (see Table A.4).

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Annex B
(informative)

Method for estimating the level effective to the ear when selecting a hearing protector based on the national peak regulation level $L'_{\mathrm{NR,peak}}$

B.1 General

This annex is applicable to all hearing protectors. At high peak sound pressure levels non-passive devices operate in the passive mode and therefore passive *HML* attenuation data are used.

A practical method for the estimation of $L'_{p,\mathrm{Cpeak}}$ is given below.

As an example, the assessment can be done by using the national peak regulation level.

Level-dependent impulse/impact noises have been classified with respect to the frequency content of the noise (see Table B.1).

The effective peak sound pressure at the ear ($L'_{p,\mathrm{Cpeak}}$) is determined using the modified sound attenuation value (d_{m}) of the hearing protector (see Table B.2). The equivalent A-weighted sound pressure level at the ear ($L'_{p,\mathrm{A,eq}}$) is determined according to Annexes A or C of this document.

B.2 Method

The following method for the estimation of a sufficient protection is based on a scientific laboratory evaluation.

Step 1: $L_{p,\mathrm{Cpeak}}$ of the noise is measured.

Step 2: Refer to Table B.1 to define the noise type.

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Table B.1 — Types of impulsive/impact noise

Noise type	Frequency range	Examples of noise sources
$L_{C,Fmax} - L_{A,Fmax} \geq 5$ dB	Where most of the acoustic energy is distributed in the lower frequency ranges	Punch press
		Piling
		Explosive (1 kg)
		Explosive (8 kg)
		Straightening steel vessel using a plastic hammer
		Large stamp press
		Jolt squeeze
		Drop forge
$L_{C,Fmax} - L_{A,Fmax} < 5$ dB	Where most of the acoustic energy is distributed between the medium and higher frequencies	Hammer on concrete
		Nail gun
		Hammer on metal
		Pistol
		Hunting rifle
		Proof firing
		Bolt firing tool

Step 3: Refer to Table B.2 to determine the d_m (dB).

Table B.2 — Modified sound attenuation value

Noise type	d_m (dB) [a]
$L_{C,Fmax} - L_{A,Fmax} \geq 5$ dB	$L - 5$
$L_{C,Fmax} - L_{A,Fmax} < 5$ dB	M
[a] Where M and L-values are obtained in accordance with EN ISO 4869-2 from passive attenuation data.	

Step 4: Calculate $L'_{p,Cpeak}$, the effective peak sound pressure level at the ear:

$$L'_{p,Cpeak} = L_{p,Cpeak} - d_m \quad \text{(B.1)}$$

Assessment: Compare $L'_{p,Cpeak}$ to the national peak regulation level ($L'_{NR,peak}$) to determine if the attenuation of the hearing protector is sufficient.

Step 5: Calculate the $L'_{p,A,eq}$

$L'_{p,A,eq}$ is determined according to Annexes A or C of this document.

Assessment: The attenuation of the hearing protector is sufficient if the assessments of $L'_{p,A,eq}$ and $L'_{p,Cpeak}$ do not exceed the national regulation levels.

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Annex C
(informative)

Selection method for sound-restoration level-dependent hearing protectors intended for all types of noise environment using *HML*-criterion values

C.1 General

This annex describes a method to aid the correct selection of sound-restoration and level-dependent hearing protectors for use in all types of noise environment. Where the user can be exposed to impulsive noise the method in Annex B should also be applied to check that the national peak regulation level $L'_{\mathrm{NR,peak}}$ has not been exceeded.

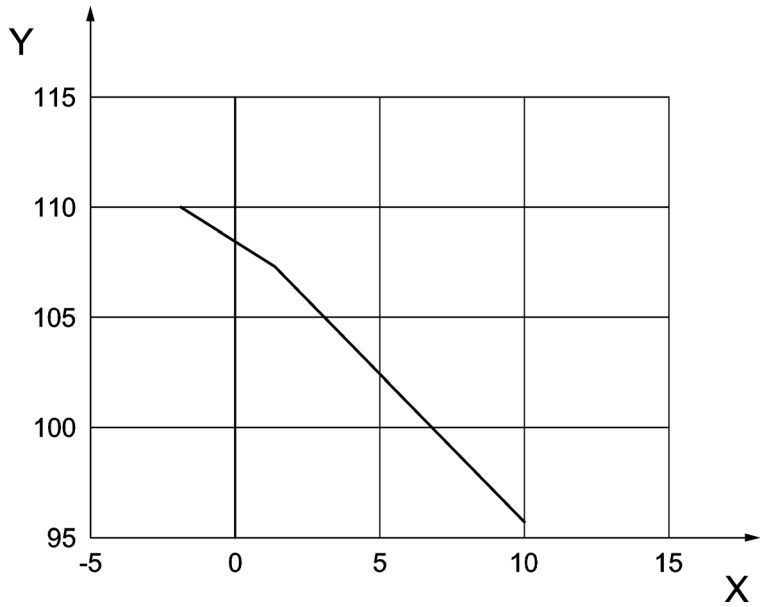
The following methods are used for checking that the A-weighted equivalent sound pressure level $L'_{p,\mathrm{A,eq}}$ effective to the ear will not exceed 85 dB(A). These methods are based on three defined criterion levels determined as per EN 352-4 or EN 352-7 as appropriate for a particular hearing protector. Three methods are given. The first and second require determination of the difference between the C- and A-weighted levels (measured as the $L_{p,\mathrm{eq}}$ of the same or equivalent events). The third requires a measurement of the A-weighted sound pressure level $L_{p,\mathrm{A,eq}}$ and an estimate of whether the sound is middle to high frequency, or low frequency.

C.2 Method 1: *HML* method

Step 1: Measure $L_{p,\mathrm{C,eq}}$ and $L_{p,\mathrm{A,eq}}$

Step 2: Calculate $L_{p,\mathrm{C,eq}} - L_{p,\mathrm{A,eq}}$

Step 3: Plot the *H*, *M* and *L* criterion levels given for the protector against corresponding $L_{p,\mathrm{C}} - L_{p,\mathrm{A}}$ values of –2 dB, 2 dB and 10 dB respectively as in the example shown in Figure C.1.



Key

x-axis: $L_{p,\mathrm{C}}$-$L_{p,\mathrm{A}}$ value in dB

y-axis: criterion level $L_{p,\mathrm{A,eq}}$ in dB

Figure C.1 — Example plot of *H*, *M*, and *L* criterion levels against the $L_{p,\mathrm{C}} - L_{p,\mathrm{A}}$ value

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Step 4: The level effective to the ear ($L'_{p,A,eq}$) will not exceed 85 dB(A) if $L_{p,A,eq}$ does not lie above the line connecting the *H*, *M* and *L* criterion levels at the calculated difference $L_{p,C,eq} - L_{p,A,eq}$ of the sound.

C.3 Method 2A: *HML* check method – Measurement check

Step 1: Measure $L_{p,C,eq}$ and $L_{p,A,eq}$.

Step 2: Calculate $L_{p,C,eq} - L_{p,A,eq}$.

Step 3: If $L_{p,C} - L_{p,A} < 5$ dB then the noise is categorized as middle to high frequency. If the $L_{p,A,eq}$ is less than or equal to the *M* criterion level, then the effective level to the ear will not exceed 85 dB(A).

NOTE The lower the value for the difference between $L_{p,C}$ and $L_{p,A}$ the greater the high frequency content of the noise.

Step 4: If $L_{p,C} - L_{p,A} \geq 5$ dB then the noise is categorized as dominant low frequency. If the $L_{p,A,eq}$ is less than or equal to the *L* criterion level, then the effective level to the ear will not exceed 85 dB(A).

C.4 Method 2B: *HML* check method – Listening method

Step 1: Measure the $L_{p,A,eq}$ of the noise.

NOTE For intermittent short-duration events it is important that the duration and the number of events included in the $L_{p,eq}$ measurement is typical of the normal working pattern.

Step 2: Decide by a listening check of the noise at the workplace and consultation of Table C.1 whether the noise is middle to high frequency, or dominant low frequency noise. Reference can also be made to the noises categorized in A.4.

Table C.1 — Examples of middle to high frequency and low frequency noise

Middle to high frequency noise	Dominant low frequency noise
Hunting rifle	Explosion
Pistol	Detonation
Proof firing	Punch press
Fireworks	Large stamp press
Nail gun	Drop forge
Impact tools	Piling
Grinder	Electric melting furnace
Drilling machine	Bulldozer
Milling machine	Excavator

Step 3: If the noise is middle to high frequency then the hearing protector is suitable if the $L_{p,A,eq}$ is less than or equal to the *M* criterion level.

Step 4: If the noise is low frequency then the hearing protector is suitable if the $L_{p,A,eq}$ is less than or equal to the *L* criterion level.

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Annex D
(informative)

Calculation method for hearing protectors with audio input

D.1 General

When selecting a hearing protector with audio input, the additional noise source represented by the audio input should be taken into account (see 5.2.4). This annex describes a method to aid the selection of such a device. These protectors are available for entertainment or for safety-related communication. This method is not applicable to level-dependent earmuffs (EN 352-4) or earplugs (EN 352-7) fitted with external audio input.

The sound pressure level effective to the ear has two components: the attenuated ambient sound and the sound from the audio input. If the audio input is for entertainment, then the hearing protectors should conform to EN 352-8 (for earmuffs) or EN 352-10 (for earplugs). For products with wireless or wired entertainment audio input these standards require the audio sound pressure level effective to the ear to be limited to 82 dB(A).

If the audio input is for safety-related communication, the hearing protectors should conform to EN 352-6 (for earmuffs) or EN 352-9 (for earplugs). Under these standards the audio sound pressure level may exceed 82 dB(A). A criterion input signal, which gives an effective level of 82 dB(A) at the ear should be provided for these unlimited protectors, according to the standards mentioned above.

D.2 Method

To ensure the daily personal exposure level effective to the ear does not exceed a national regulation level:

Estimate the level effective to the ear from the attenuated ambient noise according to Annex A. Select a hearing protector that provides an attenuated level at least 3 dB below the national regulation level.

The number of hours the audio facility can be actively used before the national regulation level is exceeded will depend on the maximum input signal to the protector.

If criterion input signal is expressed in voltage:

$$n = \mathrm{K} \cdot \left(\frac{U_{\mathrm{crit}}}{U_{\mathrm{max}}} \right)^2 \tag{D.1}$$

where

n	is the maximum number of hours for use of audio input;
U_{crit}	is the criterion voltage;
U_{max}	is the maximum rms voltage input to the protector audio input;
K	is a constant as given in Table D.1 for the corresponding national regulation level.

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If criterion input signal is expressed in dB:

$$n = \mathrm{K} \cdot 10^{\left(\frac{L_{\mathrm{crit}} - L_{\mathrm{max}}}{10}\right)} \tag{D.2}$$

where

n is the maximum number of hours for use of audio input;

L_{crit} is the criterion level in dB;

L_{max} is the maximum input signal level to the protector audio input in dB;

K is a constant as given in Table D.1 for the corresponding national regulation level.

Table D.1 shows the K values to be used for national regulation levels between 80 dB(A) and 87 dB(A).

Table D.1 — National regulation levels and corresponding K values

Regulation level [dB(A)]	K value
87	12
86	10
85	8
84	6
83	5
82	4
81	3
80	2,5

If the hearing protector has an audio system limited to 82 dB(A), as is the case for products with entertainment audio input, the value for $\left(\frac{U_{\mathrm{crit}}}{U_{\mathrm{max}}}\right)^2$ or $10^{\left(\frac{(L\mathrm{crit} - L\mathrm{max})}{10}\right)}$ equals 1. The K value in Table D.1 then corresponds to the number of hours the audio system can be used before the national regulation level is exceeded.

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Annex E
(informative)

Improving field performance and special instructions for use

E.1 Improving field performance

Experience shows that hearing protectors are often not used properly. As a consequence, they can provide less protection than anticipated.

To improve the situation, it is first and foremost to be realized that good performance does not solely come from the product itself. Good performance is most critically dependent on the user. Optimum performance relies on both careful selection and user training and motivation. These aspects become more important at higher noise levels.

Guidance for improving field performance is provided in Table E.1. Some of these are generally valid, while some are more relevant for typical groups of industrial users (and those managing them).

Table E.1 — Tips for improving field performance of hearing protectors

Attenuation	— Choose hearing protectors with an attenuation suited for the noise and work situation. If the noise exposure is distinctly different for various groups of workers (or for the same worker at different locations), then different products should be considered. — Avoid choosing protectors with the highest possible label values for people working in moderate noise levels.
Fitting and compatibility	— In principle all protectors should be individually fitted. A “one-size fits all” approach is unlikely to work well. — Conformance of individual fit is recommended for earmuffs (particularly when mounted on helmets) but is usually most critical for generic earplugs. — For earmuffs, avoid the use of spectacles with overlength and/or broad side arms as they will reduce the attenuation substantially (and often also be uncomfortable). Side arms should be thin and close to the head. Use spectacles designed for use with earmuffs.
Comfort and acceptance	— Choose a range of products that are as comfortable as possible. Discomfort is likely to discourage use of the product.
Information and training	— Awareness of the risks associated with hearing damage is essential. Information, instruction and training should be provided as appropriate (for example one-to-one or in small groups).
Individual fit testing	— Fit testing can play a valuable role in training and in achieving and documenting an effective hearing protection programme in the workplace.

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User involvement and feedback	— Involve end users in purchasing decisions. Test different products in real life and be prepared to make changes based on user feedback.
Signposting	— Clear delineation of hearing protection zones is essential.
Maintenance	— Ensure hearing protectors are regularly checked and maintained.
Management support	— Any hearing conservation program should be actively and wholeheartedly backed by management and supervisors at all levels.
Enforcement	— Individuals demonstrating unwillingness to adhere to a hearing protection programme can require redeployment into a quieter job, disciplinary action or dismissal.

E.2 Instructions for use

Information and instruction should be provided to users of hearing protection. The user should follow the instructions provided.

NOTE More frequent instruction including training is particularly important for more hazardous noise environments.

Earmuffs

When using earmuffs, during the instructions information should be given to the reduction of the protection effect by:

a) obsolete or damaged cushions;

b) use of cushions, which are damaged from storage;

c) strongly developed hair of head that could reduce the sealing between the cushion and the head;

d) earrings;

e) simultaneous use of spectacles or goggles and hearing protectors;

f) simultaneous use of a respirator mask;

g) interchange of the earmuff cups, intended for use on the right or left side of the head or top/down, if the earmuffs have a specific construction;

h) using the head band of the earmuffs not over the head but behind the neck or under the chin;

i) use of a carrier unapproved with the mounted earmuffs;

j) aging of the head band.

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Earplugs

When using earplugs the most significant problems result from incorrect handling and inserting. They can be avoided by special training.

The incorrect use results from:

1) insufficient rolling down or pressing of the user-formable earplugs;

2) insufficient deep insertion of the earplugs in the ear canal;

3) too short fixing of the inserted earplugs inside the ear canal;

4) inappropriate size of the earplugs;

5) incorrect placement of left/right-oriented earplugs;

6) using filtered earplugs without installing the filter.

All handling errors should be demonstrated by examples arising from experience in the company.

In accordance with Figures E.1 to E.4 special user training for earplugs should include:

— rolling down of the earplug;

— straightening of the ear canal;

— insertion and fixation;

— correct fitting of the earplug.

Figure E.1 — Rolling down of the earplug

It is necessary to insert the rolled or squeezed earplug in the ear canal immediately. Only in this manner is an accurate positioning with small diameter possible.

Earplugs should be inserted in the ear canal by pulling the ear upwards, thus straightening the ear canal. The straightening of the ear canal is important for insertion of all types of earplugs.

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Figure E.2 — Straightening of the ear canal

It is necessary to keep the formable earplug in place with a finger for a few seconds after insertion in the ear canal.

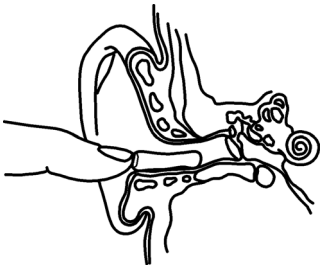

Figure E.3 — Insertion in the ear canal and fixation

The fixing is to continue until the earplug seals the ear canal (30 s or according to the manufacturer's information).

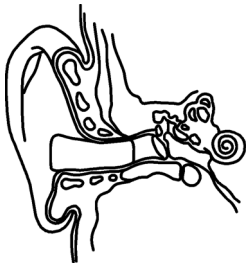

Figure E.4 — Correct fitting of the earplug

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Annex F
(informative)

Further guidance on individual fit testing methods

F.1 General information

Fit tests are a useful way to determine if selected hearing protectors provide a sufficiently good fit to achieve the anticipated sound attenuation. Fit tests can also be a helpful training aid for the user. There are a number of fit testing methods, most of which use a measured value of sound attenuation to indicate the quality of the fit. The measured value of sound attenuation can be obtained through mechanical, acoustic, subjective or objective methods. Alternatively, there are specific types of fit testing methods for a particular product or product range.

See EN 17479 for a description of fit testing systems.

F.2 Available individual fit testing methods

Currently available tests comprise:

Method 1: Sound-level measurements with microphone in real ear (MIRE)

This is an objective acoustic test method where a sound field is generated. With two microphones the sound pressure levels outside of the hearing protector and underneath the hearing protector are measured under noise load, usually simultaneously (noise reduction). With specifically determined correction factors it is possible to derive sound attenuation values in correspondence with REAT.

— Method 1a: Sound field generated by a headset.

— Method 1b: Sound field generated by a loudspeaker (free field).

Method 2: Audiometric method (determination of the threshold of hearing with and without hearing protector)

This is an acoustic test method. The thresholds of hearing with and without hearing protector are determined in a subjective measurement. The difference of the two measured thresholds of hearing is the sound attenuation.

— Method 2a: Sound field generated by a headset.

— Method 2b: Sound field generated by a loudspeaker.

Method 3: Audiometric-based method (determination of the threshold of hearing with and without hearing protector)

This is an acoustic test method. The thresholds of hearing with and without hearing protector are determined in a subjective measurement. The principle is similar to the audiometric method, but instead of an audiometer a custom-built device especially for fit testing is used.

— Method 3a: Sound field generated by a headset.

— Method 3b: Sound field generated by a loudspeaker.

NOTE The thresholds determined are not the thresholds of hearing measured by audiometric method.

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Method 4: Loudness balancing method

This is an acoustic test method. The test subject balances the loudness between the two ears with one or two earplugs and without any earplug in a subjective measurement.

Method 5: Acoustic leakage test

This is an acoustic test method. A sound pressure level is generated and measured by a loudspeaker and a microphone directly inside the earplug in the ear canal. The frequency characteristics of the sound in the ear canal contains objective information on the fitting of the earplug.

Method 6: Air leakage test

This is a non-acoustic test method, based on an air pressure measurement. The leakage of a custom moulded earplug in the ear canal is objectively determined by the decay of a small over-pressure behind the earplug or by measuring the maximum achievable over-pressure for a given maximum pressure of the pump.

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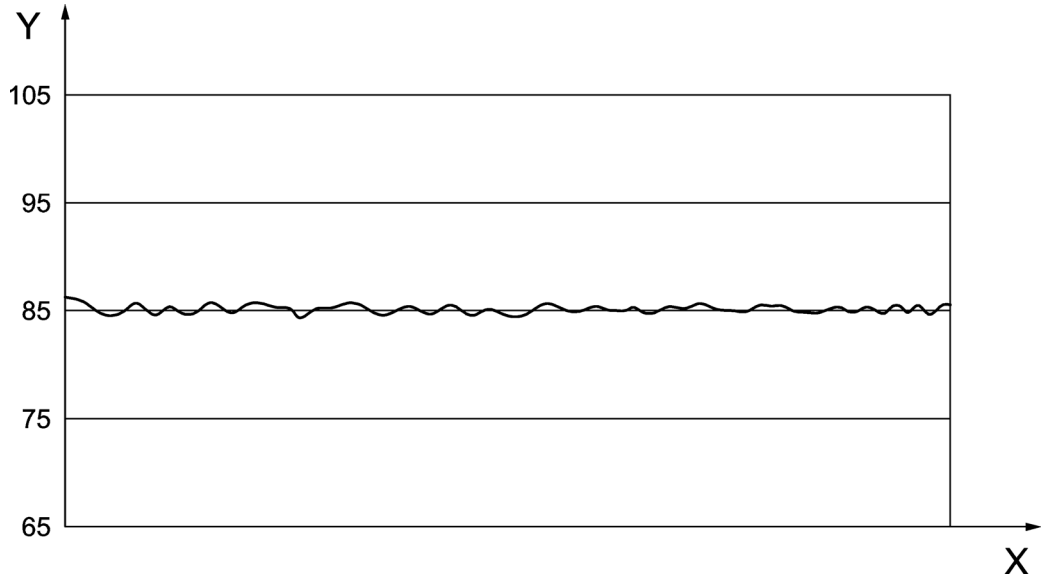
Annex G
(informative)

Noise types

Figure G.1 to Figure G.4 display the different types of noise exposure. The relation between sound pressure level and time is depicted.



Key

x-axis: time

y-axis: sound pressure level in dB(A)

Figure G.1 — Continuous noise

Continuous noise has very little variation in level over time.

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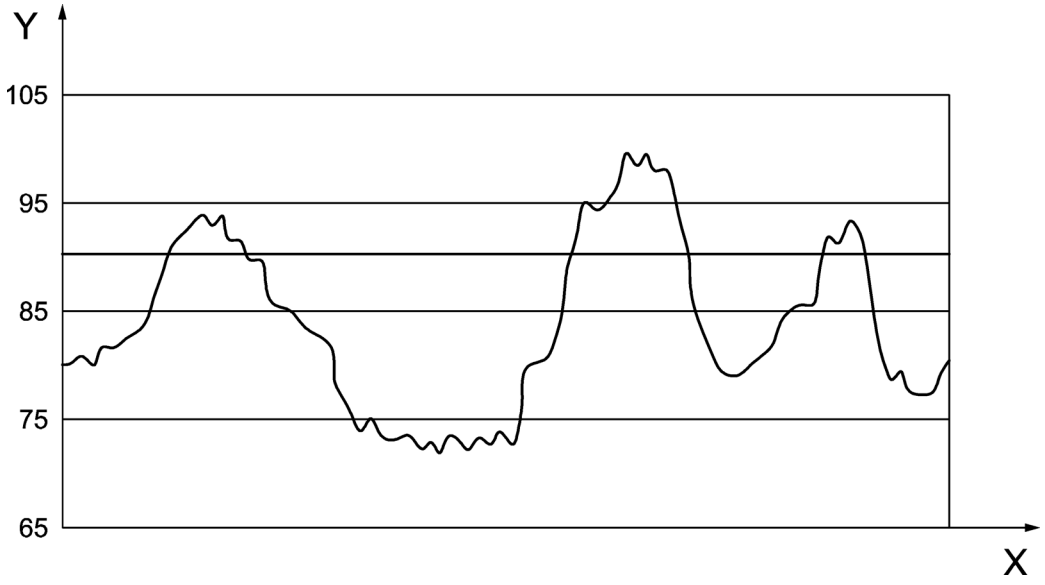


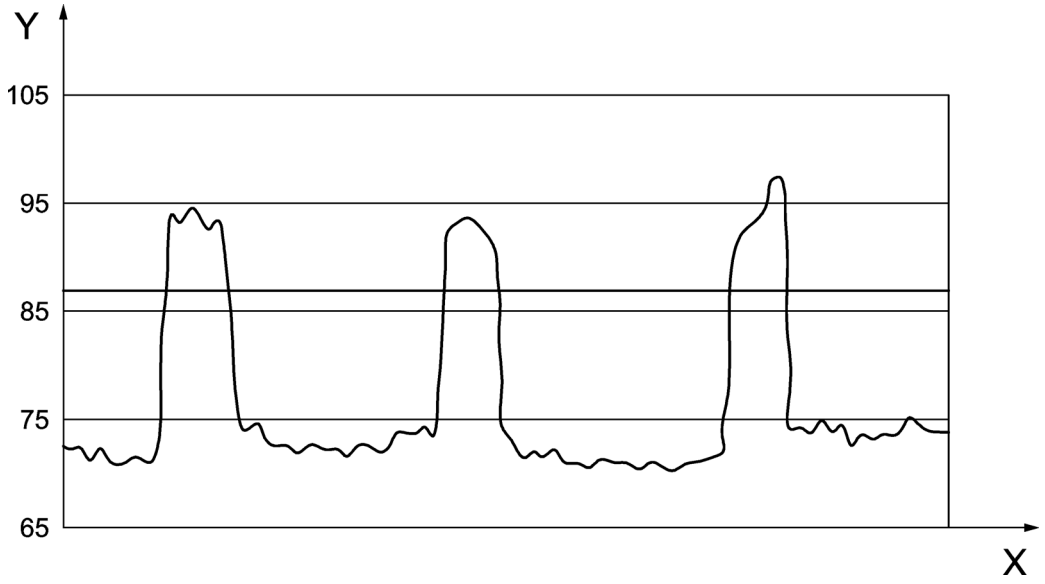
Key

x-axis: time

y-axis: sound pressure level in dB(A)

Figure G.2 — Fluctuating noise

Fluctuating noise varies irregularly in level over time.



Key

x-axis: time

y-axis: sound pressure level in dB(A)

Figure G.3 — Intermittent or repeated short-term noise

Intermittent or repeated short-term noise is similar to fluctuating noise in terms of level variation but is more predictable over time.

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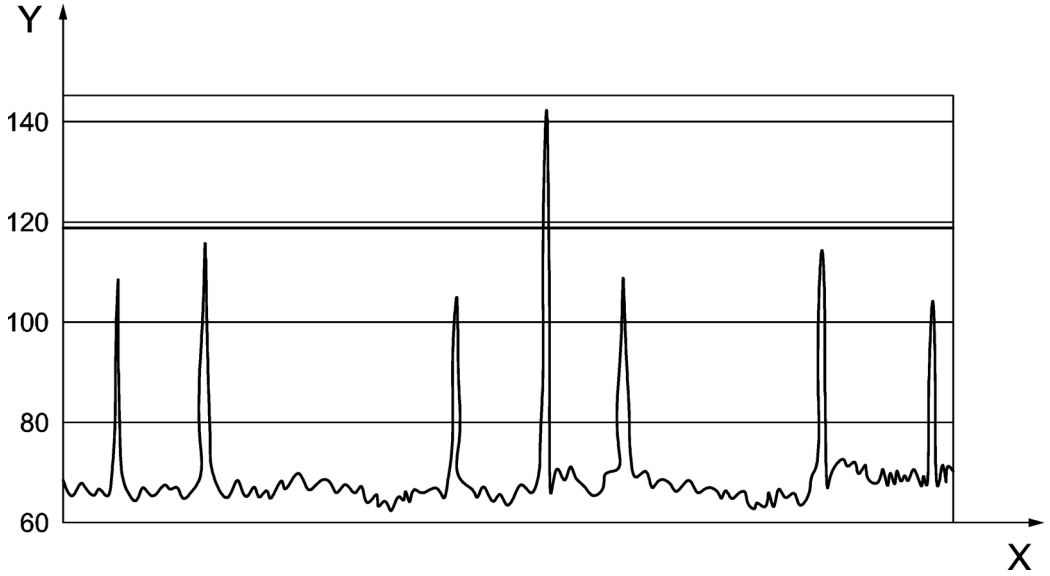

Key

x-axis: time

y-axis: peak sound pressure level in dB(C)

Figure G.4 — Impulsive noise

Impulsive noise can be characterized by short duration high-level peaks.

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Annex H
(informative)

Sustainability

H.1 General

Individuals within an organization can have different responsibilities in the selection and use of hearing protectors from the sustainability point of view.

These responsibilities include:

— sourcing (specifying certain selection criteria) (see H.2, H.3 and H.6);

— safety (checking and validating the product's suitability for the tasks, hazards and the environment) (see H.3 and H.6);

— supervision (enforcing use, checking for service and repair records, providing instruction and training on correct use) (see H.4);

— correct use (cleaning, storing, maintenance, services, disposal) (see H.5).

The following subclauses describe the steps that can be taken to improve the sustainability of hearing protectors.

H.2 Manufacturer evaluation (sourcing)

An organization, using hearing protectors, should obtain necessary information on sustainability from the manufacturers and their supply chain.

There are different ways to demonstrate compliance/engagement. Examples include, but are not limited to:

— Company level:

  — Science Based Targets initiative (SBTi);

  — B-Corporate Certification;

  — Responsible Business Alliance (RBA);

— Facility level:

  — EN ISO 14001, *Environmental management systems — Requirements with guidance for use*;

  — EN ISO 50001, *Energy Management System — Requirements with guidance for use*;

  — EN ISO 45001, *Occupational health and safety management systems — Requirements with guidance for use*;

  — ISO 26000, *Guidance on social responsibility*.

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H.3 Product attributes (sourcing and safety)

Product attributes determine the field of application. The following items should be considered [24]:

a) product durability and reliability;

b) product reusability;

c) product upgradability, repairability, maintenance and refurbishment;

d) the presence of substances of concern in products;

e) product energy and resource efficiency;

f) recycled content in products;

g) product remanufacturing and recycling;

h) product carbon and environmental footprints;

i) products' expected generation of waste materials.

The list above can be applied to the product as well as to its packaging.

NOTE 1 The applicability of certain aspects listed above can be different for passive hearing protectors and hearing protectors with added electronic functionality.

NOTE 2 All the above factors can influence the environmental impact of the products. A life cycle assessment could help to quantify the real improvement in practice.

Specific aspects for hearing protectors include, for example:

— offering reusable hearing protectors to the workforce as an alternative to disposable hearing protectors, if suitable;

— reducing the amount of waste by using bulk packaging – consult supplier for available options;

— the availability of a material recyclability programme that supports the management of disposed hearing protectors;

— avoidance of substances of public concern in hearing protectors;

— where possible, choosing hearing protectors that are supplied in sustainable and recyclable packaging;

— when selecting hearing protectors with electronic components, considering the supplier's offering of refurbished products.

H.4 Instruction and training (supervision)

The following items should be considered:

— difference between single-use and re-usable products;

— service, care, storage and maintenance of re-usable hearing protectors to prolong lifetime and retain protection performance (e.g. replacement of worn parts);

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— importance of collection of used products for material recovery or recycling;

— importance of collection of packaging material for recycling.

H.5 Cleaning, maintenance and disposal (correct use)

The following items should be considered in conjunction with the manufacturer instructions to prolong the lifetime of hearing protectors:

— for earmuffs and reusable earplugs use non-aggressive eco-friendly cleaning agents;

— for earmuffs replace cushions and inserts at an appropriate time interval;

— for earplugs with replaceable eartips change eartips at an appropriate time interval;

— routine service and repair of hearing protectors – including products featuring electronic functionality.

The manufacturer instructions on correct disposal of packaging, replaceable parts and used products should be followed.

H.6 Transportation, distribution and storage (safety and sourcing)

— Consider the impact of transportation and distribution (for example distance and method of transportation).

— Correctly store to ensure life time of the product as stated by the manufacturer.

— Establish procedures within the organization to separate materials correctly for safe recycling purpose.

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Annex I
(informative)

Significant technical changes between this document and the previous edition EN 458:2016

Table I.1 — Significant technical changes

Clause/paragraph/table figure	Technical changes
Whole document	Impact of change of hearing protectors into category III PPE according to Regulation (EU) 2016/425 considered.
Whole document	Updated in line with revision of the EN 352 series and the EN 13819 series in 2020.
Whole document	Increased relevance of fit testing taking into account EN 17479:2021.
3.12	Definition of flat frequency response to frequency range between 125 Hz and 8 000 Hz updated.
3.13 and 3.14	Definition of fit testing and related terms added.
5.1.2	Helmet mounted earmuffs renamed to earmuffs attached to head protection and/or face protection devices in line with EN 352-3.
5.2.4.2	Limiter functions mentioned explicitly.
6.2.3	Titles of clauses changed for clarification.
6.2.3.3 (now 6.2.3.4) and Annex C	Clear specification of scope of Annex C: for sound-restoration level-dependent hearing protectors not intended for use in impulse-type noise situations only.
6.2.3.6	Clear differentiation between safety-related and entertainment input.
6.3.2	Different acoustical factors and suitable hearing protectors specified in more detail.
6.3.3	Table 3 revised.
6.3.3.6 and 6.3.3.7	New environmental factors added.
6.9.1 (old clause number)	Requirements on intrinsic safety moved to 6.3.3.6.
6.9.2 (old clause number)	Requirements on electromagnetic compatibility (EMC) deleted.
7.3	Clause revised completely including fit testing with reference to EN 17479:2021. Conditions for correct fitting added.
7.8	Recommendation added to consider leisure activities together with occupational noise exposure.
9	Topic of sustainability added with reference to Annex H.
Annex A	Revised based on sound attenuation values according to EN ISO 4869-2:2018.

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Clause/paragraph/table figure	Technical changes
Annex B	Types of impulse noises reduced to two, quantitative criterium for distinction between classes added, derating for *M*-value deleted.
Annex D	Annex D deleted and content moved to 6.2.3. Following Annexes renumbered.
Annex E (now Annex D)	Calculation procedure for input signal expressed in dB added.
Annex F (now Annex E)	Information and examples on instructions for use added.
Annex G (now Annex F)	Revised based on content of EN 17479:2021.
Annex H	Information on sustainability related to hearing protection devices added in new annex.

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[5] EN 352-5, *Hearing protectors — Safety requirements — Part 5: Active noise reduction earmuffs*

[6] EN 352-7, *Hearing protectors — Safety requirements — Part 7: Level-dependent earplugs*

[7] EN 13819 (all parts), *Hearing protectors — Testing (all parts)*

[8] EN 17479:2021, *Hearing protectors — Guidance on selection of individual fit testing methods*

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